

Entity aware sequence to sequence learning using LSTMs for estimation of groundwater contamination release history and transport parameters

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Highlights

- A novel deep learning based Seq2Seq model is proposed.
- It is applied for contaminant source identification and parameter estimation.
- Entity aware LSTM is used as encoder and LSTM is used as decoder.
- EAS-LSTM takes sequential and static inputs to produce sequential outputs.
- MVO is used for inverse modelling using EAS-LSTM simulations.

Abstract

Groundwater contaminant sources identification and parameter estimation using the simulation-optimization (S/O) approach require numerous runs of the computationally expensive simulation model through the optimization algorithm. The computational cost can be effectively reduced by using a surrogate model which can accurately approximate the simulation model. With the advent of deep learning, Long Short-Term Memory (LSTM) networks, which are suitable to learn sequential data, are being increasingly applied to regression problems involving time dependencies. However, for the simultaneous contaminant source identification and parameter estimation problem, the surrogate model requires to establish a relationship between release histories at the source locations to concentration measurements at observation points subject to given values of aquifer parameters. In this study, a novel deep neural network framework Entity Aware Sequence to sequence learning using Long Short-Term Memory (EAS-LSTM) is proposed as a surrogate model which takes both sequential i.e., release histories at the source locations at different stress periods and static variables i.e., transport parameters as inputs to predict breakthrough curves (BTCs) at observation points. The proposed surrogate model is applied to a heterogeneous field scale aquifer with 4 zones. The Mean Squared Error (MSE) using EAS-LSTM is 0.0033 ppm² which is significantly better than that of Kriging and Support Vector Regression (SVR) which are 0.048 ppm² and 1.302 ppm² respectively. Comparison of EAS-LSTM model performance with Kriging and SVR based models demonstrates its higher

accuracy. Further, the optimization algorithms for inverse modelling, the performances of Multiverse Optimizer (MVO), Grey wolf optimization (GWO) and Particle swarm optimization (PSO) are investigated. It is observed that the combination of EAS-LSTM and MVO provides better results in comparison to other surrogate simulation optimization (SSO) models.

Keywords: *Contaminant source identification, Parameter estimation, Groundwater contamination, Surrogate model, Long Short-Term Memory (LSTM), Multiverse Optimization (MVO)*

1 Introduction

Groundwater resources are susceptible to contamination through various natural and anthropogenic sources (Batu, 2005). Once polluted, it is essential to reduce the contamination levels to a desirable level by considering the stakeholders involved through different remediation processes. Nevertheless, the aquifer remediation techniques are expensive and therefore they require accurate information about contamination sources such as location and release history (Sun et al., 2006). When contamination is first detected in the aquifer, the information about sources such as location and release history is unknown. This information about the sources is very crucial to predict the movement of the contaminant plume which can be used to identify the affected aquifer and predict future contamination areas which are essential for proper management of the water resources. Further, in many contamination occurrences, the hydrogeological parameters such as hydraulic conductivity, dispersivity, effective porosity etc. are unidentified, which increases the intricacy of the problem (Wagner, 1992). Moreover, the concentration measurements at observation locations are also subjected to uncertainty due to instrumental or, human errors (Yeh et al., 2007).

Mathematically, source identification and aquifer parameter estimation are inverse problems. Amongst various approaches, the simulation-optimization (S/O) approach is widely applied to solve this problem (Aral et al., 2001; Yeh et al., 2007; Ayvaz, 2010; 2016; Gurarslan and Karahan 2015). This approach has also been used for the estimation of parameters and source identification (Wagner 1992; Mahar and Datta, 2001). In this approach, the optimization technique employs a simulation model to evaluate several combinations of release rates at possible source locations and aquifer parameters to produce agreeable concentrations at observation points. Nonetheless, this approach is computationally expensive as the simulation model, which solves the complex boundary value problem of groundwater flow and transport, is summoned numerous times by the optimization algorithm (Li et al., 2020). Alternatively, these simulation

models can be replaced with surrogate models which are trained using datasets prepared from the former. A well-trained surrogate model can efficiently approximate the simulation models and reduce the computational complexity.

Surrogate models such as Feed Forward Neural Networks (FFNN) (Majumder and Eldho, 2020; Singh and Datta, 2004), Extreme Machine Learning (Li et al., 2020; Yadav et al., 2016) and Kriging (Zhao et al., 2016) have been successfully implemented for simulating complex hydrological and hydrogeological environment. Although these methods have been proven effective, these techniques are not inherently designed for learning the data in the form of multiple sequences. Therefore, the analyzing capability of the temporal dependencies between the sequential input and output data is restricted in these models. Recurrent Neural Network (RNN), which is a type of Deep Neural Network (DNN), is specifically designed for learning these dependencies between input and output. Long Short-term Memory (LSTM) is a variant of RNN which was proposed by Hochreiter and Schmidhuber (1997) to overcome the exploding and vanishing gradient problem in simplistic or, vanilla RNNs. In LSTM, the input data is transformed into an internal state sequentially which is later deciphered to return output in a sequential form. It has been successfully used in many applications in water resources such as streamflow prediction (Gao et al., 2020), water table depth forecasting (Zhang et al., 2018), water quality modelling (Liang et al., 2020) and flood forecasting (Lv et al., 2020). However, these types of LSTM models are suitable where input and output sequences are of the same length. For problems where the lengths of input and output sequences are different, a lately developed strategy i.e., encoder-decoder LSTM (ED-LSTM) is applied (Sutskever et al., 2014; Kao et al., 2020). In this sequence-to-sequence (Seq2Seq) learning approach, the input sequence is initially encoded into internal states using an LSTM which is later decoded using another LSTM.

The present study aims to develop a DNN based surrogate model which maps release rates at possible source locations and transport parameters to breakthrough curves (BTCs) at the observation points. Here release rates and BTCs are input and output sequences respectively. In contrast, transport parameters are static and specific to the study area. Therefore, the current study can be deemed as entity aware Seq2Seq learning. As LSTMs necessitates both input and output data to be in sequential form, a new type of LSTM i.e., Entity Aware LSTM (EALSTM) was proposed by Kratzert et al. (2019) to handle both static and dynamic/ sequential data as input. Our study proposes EALSTM to be used as an encoder in the Seq2Seq learning. It transforms the release rates at possible source locations and transport parameters into an encoded state. Subsequently, a decoder LSTM is used to derive the output sequence from the encoded state in a

recursive manner. The proposed model is named Entity Aware Sequence to sequence learning using Long Short-Term Memory (EAS-LSTM). The training data set is prepared using inputs from meshfree Radial Point Collocation Method (RPCM) simulation model which is advantageous compared to standard grid/mesh-based methods such as Finite Difference Method (FDM) and Finite Element Method (FEM) in terms of simplicity of the model, ease of domain discretization, handling high advection and reaction rates for contaminant transport problems etc. (Liu and Gu, 2005; Singh et al., 2016; Anshuman et al., 2019; Anshuman and Eldho, 2020). Also, the use of a meshfree method facilitates the addition/deletion of nodes without altering the node structure for the entire domain which is helpful in adaptive analysis (Pathania et al., 2020). RPCM model has been successfully used for groundwater problems such as flow simulation (Meenal and Eldho, 2011), contaminant transport simulation (Anshuman et al., 2019) and coupled flow-transport simulation (Guneshwor et al., 2018; Anshuman and Eldho, 2020).

In inverse modelling using the S/O approach, the optimization model also plays a significant role. Some of the popular optimization algorithms in groundwater-related problems are Genetic Algorithm, Particle Swarm Optimization (PSO), Cat Swarm Optimization, Grey Wolf Optimization (GWO) and Differential Evolution (DE) (Singh and Datta, 2006; Thomas et al., 2018; Guneshwor et al., 2018; Majumder and Eldho, 2020). These algorithms try to achieve an optimal solution through a search process that can be broadly divided into exploration and exploitation phases. In the exploration phase, an optimization algorithm looks for promising regions of optimum solution in the global search domain. While in the latter, it tries to converge to global optimum around the obtained promising regions. A recently proposed Multiverse Optimization (MVO) algorithm ensures a balance between exploration and exploitation phase by providing smooth transition from former to latter (Mirjalili et al., 2016). In this study, MVO is used as an optimization tool for the identification of contamination sources and transport parameters through EAS-LSTM surrogate model simulations for a case study. The performance of the proposed surrogate model is compared with that of widely used Kriging and Support Vector Regression (SVR) based surrogate models. Moreover, the performance of MVO is studied against that of GWO and PSO algorithms.

2 Methodology

2.1 Groundwater flow and transport simulation

The steady-state flow in a heterogeneous and anisotropic unconfined aquifer is governed by Eq. (1) (Bear and Cheng, 2010; Rastogi, 2007)

$$\frac{\partial}{\partial x_i} \left(\kappa_{x_i} h \frac{\partial h}{\partial x_i} \right) = -Q \quad (1)$$

Here, i represents the spatial dimensions of the problem domain, x_i represents the direction along the i^{th} dimension. The hydraulic conductivity and piezometric head are represented by κ and h respectively. The term Q represents the source/sink term present in the aquifer domain. The boundary of the problem domain is subjected to Dirichlet and Neumann boundary conditions given by Eq. (2) and (3), respectively.

$$h(x_i) = h_o \quad (2)$$

$$\kappa_{x_i} \frac{\partial h}{\partial n_{x_i}} = f_o \quad (3)$$

where h_o and f_o are known head and flux values respectively. The direction normal to the boundary is represented by n_{x_i} .

The transport process of single reactive species in an aquifer is governed by (Bear and Verruijt, 2012)

$$\frac{\partial c}{\partial t} = \frac{\partial}{\partial x_i} \left(D_{x_i x_j} \frac{\partial c}{\partial x_j} \right) - \frac{\partial}{\partial x_i} (v_{x_i} c) - \lambda c + Q_c \quad (4)$$

where j is another spatial dimension, c is the concentration of the species, D is the dispersion tensor, v is Darcy velocity, Q_c is the release rate and λ is the first-order reaction rate. In the coupled flow and transport system, the Darcy velocity is obtained using hydraulic conductivity (κ), effective porosity (η_e) and head distribution (estimated by solving Eq. (1-3)) as follows.

$$v_{x_i} = -\frac{\kappa}{\eta_e} \frac{\partial h}{\partial x_i} \quad (5)$$

In a 2D domain, the dispersion tensor is estimated by (Bear and Cheng, 2010)

$$D_{x_1 x_1} = \frac{\alpha_L v_{x_1}^2 + \alpha_T v_{x_2}^2}{\sqrt{v_{x_1}^2 + v_{x_2}^2}}, D_{x_2 x_2} = \frac{\alpha_T v_{x_1}^2 + \alpha_L v_{x_2}^2}{\sqrt{v_{x_1}^2 + v_{x_2}^2}} \quad (6)$$

Here α_L and α_T represent longitudinal and transverse dispersivities respectively. The Dirichlet and Neumann type conditions for contaminant transport are given by Eq. (7) and (8) respectively.

$$c(x_i) = c_o \quad (7)$$

$$D_{x_i x_j} \frac{\partial c}{\partial n_{x_i}} = g_o \quad (8)$$

where c_o and g_o are applicable known concentrations and fluxes respectively.

In this study, the simulation is done using the meshfree radial point collocation method (RPCM) which has been successfully used for groundwater modelling (Meenal and Eldho, 2011; Guneshwor et al.,

2018; Anshuman et al., 2019; Anshuman and Eldho, 2020). In RPCM, the problem domain is decomposed into several overlapping support domains which surround each node (See Fig. 1).

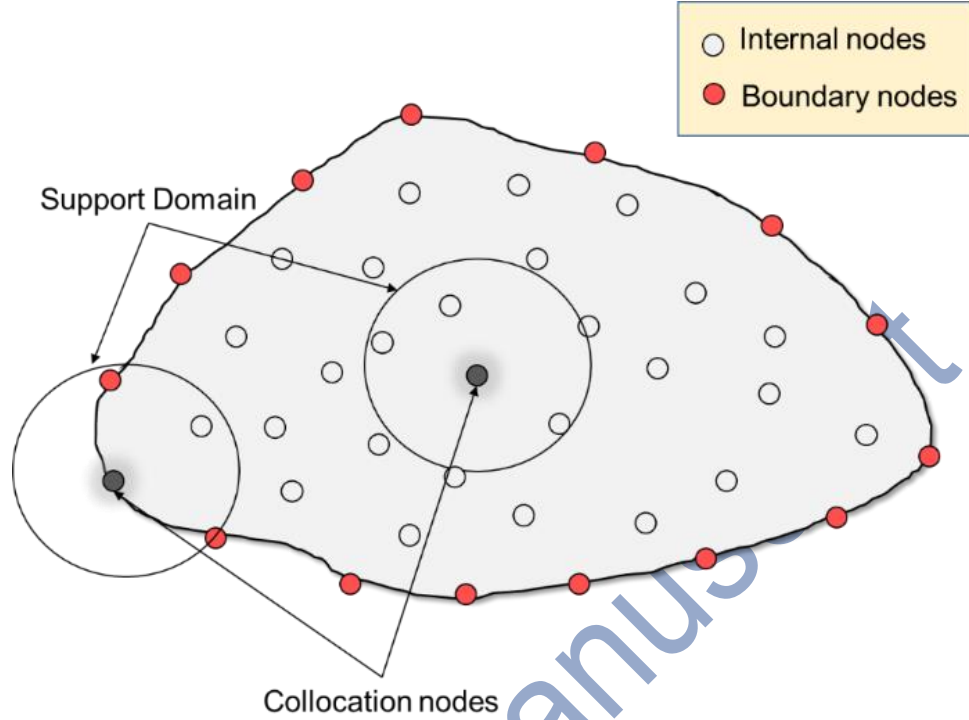


Fig.1. Representation of spatial discretization of problem domain using RPCM

A variable ‘ h ’ at location x_i can be approximated using n nodes in its support domain as shown in Eq. (9) (Liu and Gu, 2005).

$$h(x_i) = \sum_{k=1}^n f_{R_k}(x_i, x_k) \cdot a_k \quad (9)$$

Here, f_{R_k} stands for multi-quadrics radial basis function (MQ-RBF) evaluated for k^{th} node in the local support domain which is given by (Kansa, 1990a; 1990b)

$$f_R(x_i, x_k) = [(\|x - x_k\|)^2 + (\alpha \cdot d_c)^2]^q \quad (10)$$

The term $\|x - x_k\|$ represents the Euclidean distance between node x_i and the point of interest x_k and d_c is the average nodal distance considering all internal nodes. The MQ-RBF shape parameters are denoted by q and α . The unknown coefficients a_k in Eq. (9) can be estimated by enforcing the MQ-RBF moment matrix to pass through the nodes in the local support domain as follows (Liu and Gu, 2005).

$$\{a\}_{k \times 1} = \begin{bmatrix} f_R(x_1, x_1) & \dots & f_R(x_1, x_k) \\ \vdots & \ddots & \vdots \\ f_R(x_k, x_1) & \dots & f_R(x_k, x_k) \end{bmatrix}^{-1} \cdot \{\hat{h}\}_{k \times 1} = G^{-1} \cdot \{\hat{h}\}_{k \times 1} \quad (11)$$

where $\hat{h}$ is the approximated solution of the variable of h . By substituting Eq. (11) in Eq. (9) the solution for h can be obtained as follows (Liu and Gu, 2005)

$$h(x_i) = \{f_R\}' \cdot [G]^{-1} \cdot \{\hat{h}\} = \{\phi\}\{\hat{h}\} \quad (12)$$

The term ϕ is referred to as shape function which is estimated for each node within its local support domain. The derivatives of the shape functions concerning directions can be estimated by swapping the MQ-RBF vector (f_R) in Eq. (12) with the corresponding derivative terms. Hence, the RPCM solutions to the governing equation (1) can be represented in terms of the shape functions and their derivatives in Eq. (13). It should be noted that as the concerning PDE is nonlinear with respect to h , it is solved iteratively (Meenal and Eldho, 2011).

$$\left[\left(\frac{\partial \phi}{\partial x_i} \cdot \kappa_{x_i} \right) \cdot h \cdot \frac{\partial \phi}{\partial x_i} + \kappa_{x_i} \cdot \frac{\partial \phi}{\partial x_i} \cdot h \cdot \frac{\partial \phi}{\partial x_i} + \kappa_{x_i} \phi \cdot \frac{\partial^2 \phi}{\partial x_i^2} \right] \cdot h = Q. \quad (13)$$

Likewise, the governing equation for contaminant transport Eq. (4) can be discretized as follows

$$C^{t+1} - (1 - \theta)\Delta t \left[D_{x_i x_i} \frac{\partial^2 \phi}{\partial x_i^2} - v_{x_i} \frac{\partial \phi}{\partial x_i} - \lambda \phi \right] \cdot C^{t+1} = C^t + \theta \cdot \Delta t \left[D_{x_i x_i} \frac{\partial^2 \phi}{\partial x_i^2} - v_{x_i} \frac{\partial \phi}{\partial x_i} - \lambda \phi \right] \cdot C^t \quad (14)$$

A semi-implicit scheme is employed for the temporal discretization of concentration in the governing equation Eq. (4) (Anshuman et al., 2019). The scheme is referred to as implicit if $\theta=0$ and explicit if $\theta=1$. In this study, the Crank-Nicolson scheme is employed by fixing the parameter θ as 0.5 (Rastogi, 2007) as this scheme gives accurate transient simulations (Anshuman and Eldho, 2020). A detailed study on sensitivity analysis for the parameter θ is given in Anshuman and Eldho, (2020).

2.2 A brief overview of LSTM

Artificial neural networks (ANN) such as Feed Forward Neural Networks (FFNN) has been used in the past due to their simplicity and high accuracy (Majumder and Eldho, 2020; Singh and Datta, 2004). However, in FFNN the time dependency in the input data is not learnt sequentially as the training data is flattened and the sequential/temporal information is lost. Conversely, in Recurrent Neural Networks (RNN), the information is allowed to be carried over different time steps which are beneficial for learning the sequential dependency in the input data making it more suitable even for long time-series (Kao et al., 2020). Neural networks are generally trained using the Back Propagation technique using gradient descent algorithms. Nevertheless, RNNs suffer from *vanishing* and *exploding* gradients issues increasingly with the time steps which aggravates the model training process.

This issue can be effectively dealt with by using Long Short-Term Memory (LSTM) (Hochreiter and Schmidhuber, 1997) which is a special type of RNN. It uses a combination of three gates namely input,

forget and output (See Fig. 2 a). These gates operate on the given input (I) and hidden state (H) at time step “ t ” using sigmoid activation function (σ) as shown in Eqs. (15-17).

$$f_t = \sigma(w_{1f}I^t + w_{2f}H^{t-1} + b_f) \quad (15)$$

$$i_t = \sigma(w_{1i}I^t + w_{2i}H^{t-1} + b_i) \quad (16)$$

$$o_t = \sigma(w_{1o}I^t + w_{2o}H^{t-1} + b_o) \quad (17)$$

Here f_t , i_t and o_t are sigmoid activation vectors obtained from forget, input and output gates respectively. The respective terms w_1 and w_2 are weights associated with input and hidden state whereas b represents the bias vector. The subscripts f , i and o correspond to forget, input and output gates respectively. The cell state C_t is computed using antecedent cell state vector $\bar{C}_t$ which is a function of input at time t and hidden state vector of the previous time step (Eq. (18) and (19)). Similarly, hidden state H_t is computed using o_t and cell state C_t (Eq. (20)). The notation $\odot$ refers to Hadamard or, the dot product between concerning vectors.

$$\bar{C}_t = \tanh(w_{1c}I^t + w_{2c}H^{t-1} + b_c) \quad (18)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \bar{C}_t \quad (19)$$

$$H_t = o_t \odot \tanh(C_t) \quad (20)$$

2.3 Entity aware LSTM and sequence-to-sequence learning

As discussed in the introduction section, the objective of the study is to develop a DNN which learns the relationship between the contaminant release history at source locations and observed concentrations at observation points subject to certain values of transport parameters. The release history and contaminant observation data are sequential or, dynamic. On the other hand, the transport parameters such as dispersivity (α_L) and effective porosity (η_e) are dependent on the soil type and are static. As LSTM requires both input and output data in sequential form, a modified version of LSTM i.e., Entity Aware Long Short-Term Memory (EALSTM) which can take both static and sequential variables as input is proposed for this study (Kratzert et al., 2019). In EALSTM, the conventional LSTM is modified by introducing the static variables in the input gate (See Fig. 2b). Mathematically, the gate operations in LSTM i.e., Eqs. (15-20) are modified as follows (Kratzert et al., 2019).

$$f_t = \sigma(w_{1f}I_d^t + w_{2f}H^{t-1} + b_f) \quad (21)$$

$$i = \sigma(w_{1i}I_{st} + b_i) \quad (22)$$

$$o_t = \sigma(w_{1o}I_d^t + w_{2o}H^{t-1} + b_o) \quad (23)$$

$$\bar{C}_t = \tanh(w_{1c}I_d^t + w_{2c}H^{t-1} + b_c) \quad (24)$$

$$C_t = f_t \odot C_{t-1} + i \odot \bar{C}_t \quad (25)$$

$$H_t = o_t \odot \tanh(C_t) \quad (26)$$

Here I_d and I_{st} are dynamic/ sequential and static inputs respectively. Note that the f and o gates are computed at every timestep using I_d while the input gate (i) is evaluated only once using I_{st} .

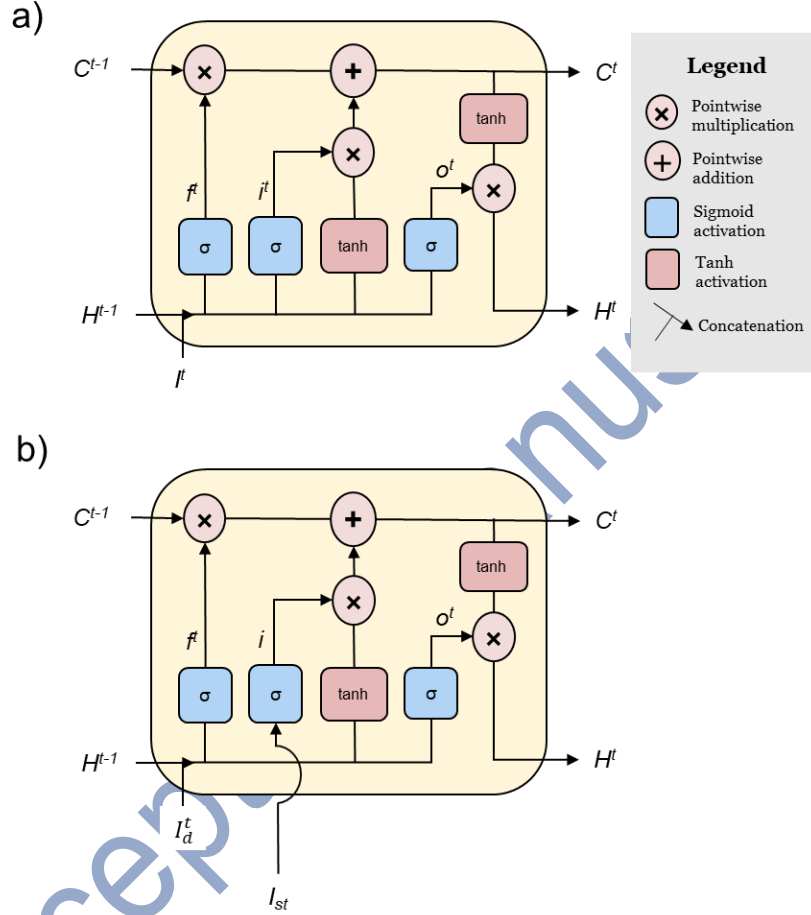


Fig. 2. Schematic representation of a) an LSTM cell and b) an Entity Aware LSTM cell

As shown in Fig. 2, in both LSTM and EALSTM each memory unit outputs a hidden state (H^t) and cell state (C^t) at each timestep. The number of neurons used in the hidden states is generally higher than that of the input sequence (Zhang et al., 2018; Gao et al., 2020). A fully connected layer is further used to transform these states to output (Zhang et al., 2018). However, this approach is not suitable when the lengths of input and output sequences are unequal as in the current study (Kao et al., 2020). Hence, a sequence-to-sequence (Seq2Seq) learning strategy is employed which uses an encoder-decoder technique to map the input sequence to output sequence (Kao et al., 2020; Xiang et al., 2020). In this framework, the input data is sequentially fed into the encoder. In this study, EALSTM is proposed as the encoder. As shown in Fig.

2, EALSTM encodes the input into hidden and cell states at each timestep. The encoded hidden and cell states after the last timestep are set as initial states for the decoder LSTM which generates the output hidden sequences recursively. Using a fully connected layer, these hidden sequences are transformed into the output sequence. This DNN architecture can accommodate input and output sequences of different lengths which is suitable for source identification problems as the stress periods of contaminant releases and measurements in observation wells can be of unequal lengths. Further for the current study, the DNN is required to learn the input and output sequences for a given set of transport parameters i.e., zonal dispersivities and porosities which are static variables. The proposed encoder-decoder model for the study is named Entity Aware Sequence-to-sequence learning using Long Short-Term Memory (EAS-LSTM). The architecture of the model is provided in Fig. 3. As observed, the encoder takes both static and dynamic/ sequential variables. The input gate i.e., Eq. (22) is evaluated only once and re-used by EALSTM at every timestep to evaluate corresponding hidden and cell states. Here it may be noted that a proxy input for the first timestep is provided at the decoder LSTM for the first timestep. The proxy input consists of a zero matrix as the input dataset used to train the EAS-LSTM model are mutually exclusive. Hence, decoder LSTM is forced to learn the first timestep using encoded states from encoder EALSTM only.

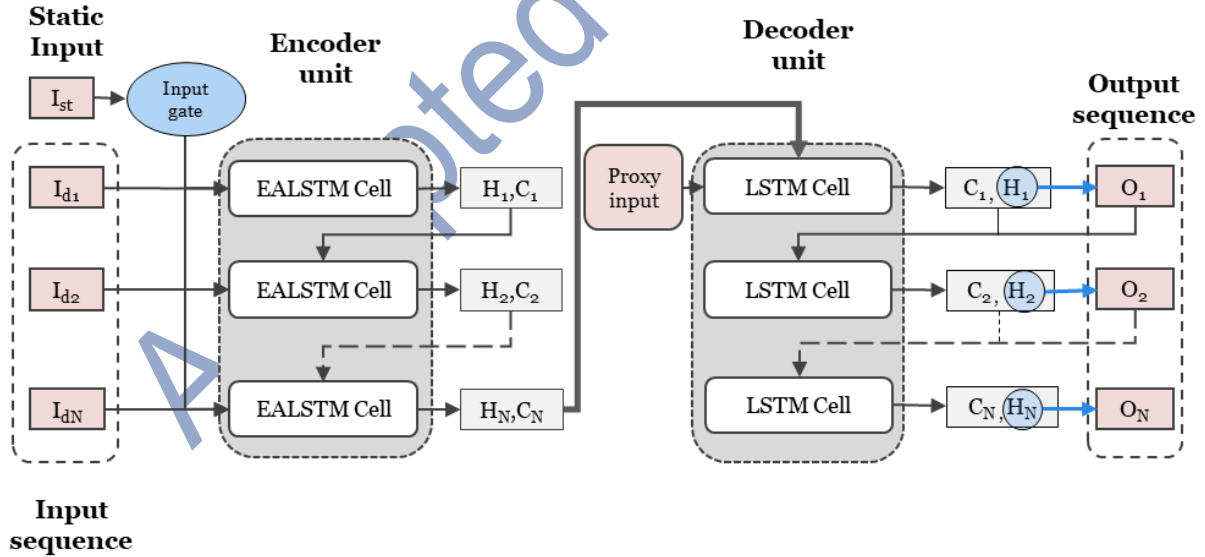


Fig. 3 EAS-LSTM model architecture using EALSTM as encoder and LSTM as the decoder. The blue arrows from hidden states of the decoder denote the transformation of hidden states to outputs using fully connected layers

In the present study, the dynamic/sequential component of the input data i.e., release rates from source locations are arranged in matrix form as $n_{sp} \times n_{source}$ where n_{sp} and n_{source} stand for the number of stress periods and source locations respectively. On the other hand, the static component i.e., zonal transport parameters are arranged in a vector form $n_T \times 1$ where n_T is the number of transport parameters. Similar to sequential input data, the output data is arranged in $n_t \times n_{obs}$ where n_t and n_{obs} are the number of timesteps and observation points respectively. The model setup is presented in simplified form in Fig. 4.

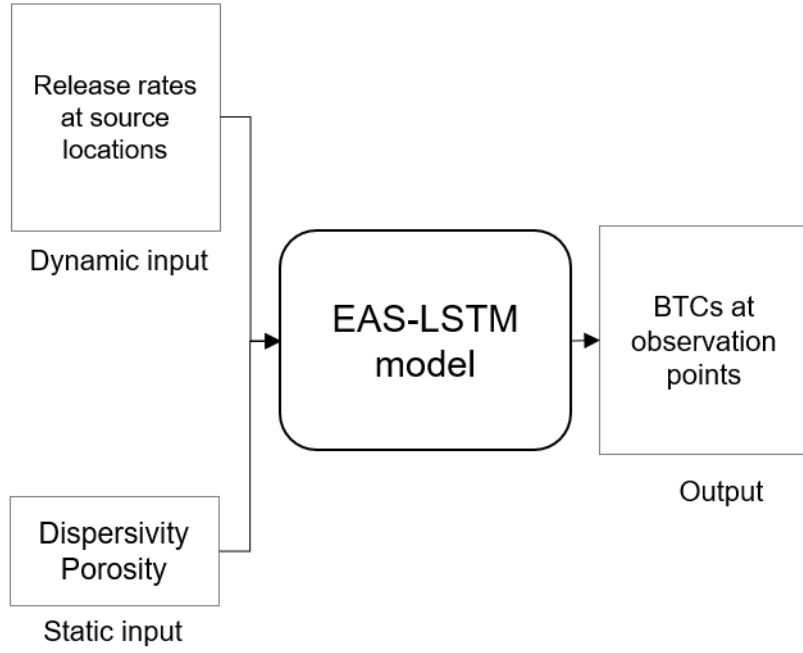


Fig. 4. Simplified representation of EAS-LSTM model setup

2.4 Kriging

Kriging is a geostatistical approach in which the interpolated values are estimated by a Gaussian process based on prior covariances (Matheron, 1963). It has been applied in groundwater problems such as contaminant source identification (Zhao et al., 2016), monitoring of groundwater monitoring network (Dhar and Datta, 2010) and prediction of groundwater depth (Nikroo et al., 2010). The predicted output $\hat{y}$ is estimated by a linear combination of weighted deterministic function $f(x)$ and realization of a stochastic process $\xi(x)$ as follows

$$\hat{y} = \sum_{i=1}^k \beta_i f_i(x) + \xi(x) \quad (27)$$

Here the mean of the realization $\xi(x)$ is zero. Its covariance is estimated as follows

$$Cov[Z(x_i), Z(x_j)] = \sigma^2 R \quad i, j \in [1, N] \quad (28)$$

Here σ^2 , R and N are process variance, Gaussian correlation function and the number of data points. More details on the Kriging method are provided in Zhao et al., 2016. In the present study, the release rates and transport parameters are provided as the input vector. Similarly, the BTC curves at observation wells are also obtained in the form of a vector.

2.5 Support Vector Regression

Support Vector Regression (SVR) is a variant of support vector machine (SVM) that tries to fit the input data to a hyperplane with some acceptable error. SVR has been used for contaminant source identification (Xing et al., 2019; Pan et al., 2021) and groundwater potential mapping (Panahi et al., 2020). In SVR, the model prediction $\hat{y}$ for input, x is represented as follows.

$$\hat{y} = \omega^T \cdot \phi(x) + \varepsilon \quad (29)$$

Where ω is the coefficient vector, ϕ is the kernel function and ε is the bias. In this study, a commonly used radial basis function (RBF) kernel is applied (Al-Fugara et al., 2020). The unknowns ω and ε are estimated by solving the constrained minimization problem given by

$$\text{Minimize } \frac{1}{2} \|\omega\|^2 + C \sum_{i=1}^N |\xi_i| \quad (30)$$

such that

$$\begin{cases} |y_i - \omega_i x_i| \leq \varepsilon + |\xi_i| \\ \xi_i \geq 0 \end{cases} \quad i \in Z : 1 \leq i \leq N \quad (31)$$

Here $\|\omega\|$ is the L2 norm of the coefficient matrix, C is the regularization parameter and ξ is the relaxation parameter. The term Z and N represent integer and number of data points respectively. The optimization problem can be solved using Lagrange multipliers as given in Xing et al., (2019). The input and output data for the SVR surrogate model are arranged in a similar way as explained in the previous section.

2.6 Optimization algorithms

2.6.1 Multiverse optimization algorithm

Multi-Verse Optimization (MVO) is a recently proposed stochastic population-based optimizer (Mirjalili et al., 2016). According to a popular theory in physics, there exists a multiverse consisting of innumerable universes each created by a big bang (Tegmark, 2004). In this algorithm, the candidate solutions are referred to as universes. These universes intermingle by exchanging components using white holes, black holes and wormholes. The black holes are observed throughout the universe and are known to devour anything close enough due to their immense gravitational force. In contrast white holes eject matter into space and are

theorized to be created due to a collision between universes (Eardley, 1974). The wormholes are theorized as cosmic tunnels connecting different parts of space-time even the multiverse (Morris and Thorne, 1988).

In MVO, it is assumed that only one of the universes, i.e., the best universe, contains a white hole while the others contain black holes (Mirjalili et al., 2016). The objects are sent from the universe having a white hole to other universes through white hole-black hole tunnels established by the roulette wheel mechanism (Mirjalili et al., 2016).

$$x_i^j = \begin{cases} x_k^j, & \text{if } r_1 < NI_i \\ x_i^j, & \text{otherwise} \end{cases} \quad (32)$$

Here x_i^j is j^{th} variable and NI_i is the normalised inflation rate for of i^{th} universe respectively and x_k^j denotes j^{th} variable of k^{th} universe selected through roulette wheel mechanism. The term r_1 is the random number in $[0, 1]$. Each universe is assigned an inflation rate that is proportional to the fitness value. The best solution after each iteration is considered to the universe with the best inflation/expansion rate. In MVO, the interaction between universes follows the following principles (Mirjalili et al., 2016)

1. A higher inflation rate denotes a higher probability of having a white hole.
2. A higher inflation rate denotes a lower probability of having a black hole.
3. The objects are sent through a white hole from the universe having a higher inflation rate.
4. The objects are received through black holes through the universes having lower inflation rates.
5. The objects from all universes may move towards the best universe through the wormholes irrespective of the inflation rates.

From the above rules, it is established that the exploration of the search space is done using white holes/black holes while the exploitation or, local search is done using wormholes. Every universe can possess a wormhole. The local changes in the universes through wormholes is accomplished by Eq. (33) (Mirjalili et al., 2016).

$$x_i^j = \begin{cases} \begin{cases} x_{best}^j + TDR \times [(ub_j - lb_j) \times r_4 + lb_j], & \text{if } r_3 > 0.5 \\ x_{best}^j - TDR \times [(ub_j - lb_j) \times r_4 + lb_j], & \text{otherwise} \end{cases} & \text{if } r_2 < WEP \\ x_i^j & \text{otherwise} \end{cases} \quad (33)$$

Where x_{best}^j is the j^{th} variable of the best universe; ub_j and lb_j are upper bounds and lower bounds of the j^{th} variable and r_2, r_3 and r_4 are random numbers in $[0,1]$. The terms WEP and TDR represent wormhole existence probability and travelling distance rate respectively which are explained in Eq. (34) and (35) (Mirjalili et al., 2016).

$$WEP_t = WEP_{min} + t \times \frac{WEP_{max} - WEP_{min}}{t_{max}} \quad (34)$$

$$TDR = 1 - \left(\frac{t}{t_{max}} \right)^{1/n} \quad (35)$$

Here t and t_{max} are the iteration number and maximum iteration number respectively. It is observed that the WEP increases linearly with the iteration number within the limits $[WEP_{min}, WEP_{max}]$. From Eq. (34) it is observed that the wormhole is only opened if $r_2 < WEP$. Since WEP increases linearly with the iteration number, the chances of local search or exploitation increase and global search or, exploration decreases with an increase in iteration number. This essentially means that the MVO algorithm smoothly transforms from the dominance of the exploration phase to that of the exploitation phase. In this study, WEP_{max} and WEP_{min} are fixed as 1 and 0.2 respectively. The exponent $\frac{1}{n}$ in Eq. (35) introduces non-linearity to TDR with respect to iteration number. In this study, the value of n is fixed as 6 which is suggested by Mirjalili et al., (2016). Higher values of n result in more accurate exploitation (Sayed et al., 2019). The steps involved in MVO is presented in Fig. 5.

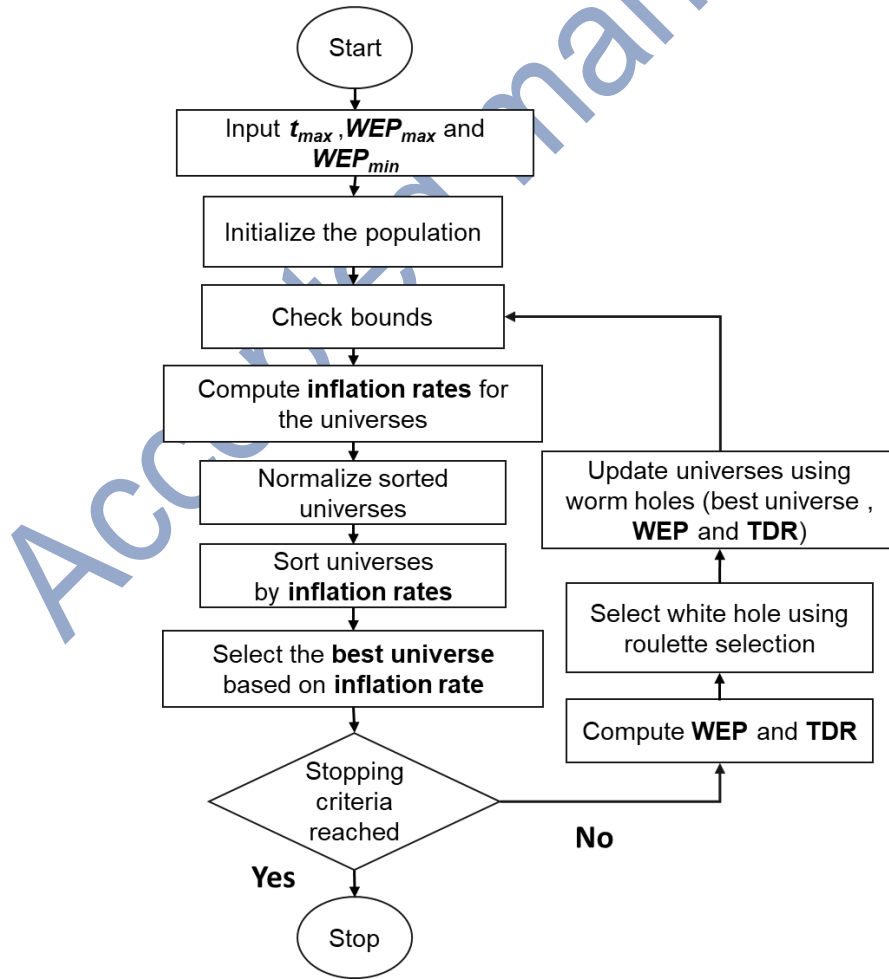


Fig. 5. Flowchart depicting the steps involved in MVO

2.6.2 Grey wolf optimization

GWO is an optimization algorithm inspired by the hunting behaviour of Grey Wolves (Mirjalili et al., 2014). Based on leadership and dominance, the wolves are categorized into alpha, beta, delta and omega. The first three categories lead the pack for hunting while omega wolves hound the prey. Alpha is the most dominant category followed by beta and delta. Mathematically, the encircling of prey by the grey wolf pack is denoted as follows:

$$D = |C \times x_w^t - x_i^t| \quad (36)$$

$$x_i^{t+1} = x_w^t - A \times D \quad (37)$$

The candidate solution is represented by the position of wolves in the pack i.e., x^t where t denotes the iteration number. x_w^t represents the position of the leading wolf (alpha, beta or, delta). A and C are the coefficients based on uniformly distributed random numbers. Further details of the GWO algorithm can be found in Mirjalili et al. (2014) and Majumder and Eldho (2020).

2.6.3 Particle swarm optimization

PSO is another stochastic population-based algorithm that mimics the movement of a bird flock (Kennedy and Eberhart, 1995). Here the position of each particle represents a candidate solution. All particles possess some velocity which is updated in each iteration using which they traverse the search space. The particle velocities are updated at the end of each iteration as follows:

$$v_i^{t+1} = \omega \times v_i^t + C_1 \times r_1 \times (p_{best} - x_i^t) + C_2 \times r_2 \times (g_{best} - x_i^t) \quad (38)$$

Here ω is the inertia coefficient and C_1 and C_2 are acceleration coefficients. The terms r_1 and r_2 are the random numbers in the interval $[0, 1]$. The term p_{best} and g_{best} denote best positions of the particle and swarm respectively. The positions of particles are updated after each iteration by

$$x_i^{t+1} = x_i^t + v_i^t \quad (39)$$

3 Problem statement and modelling framework

3.1 Study area and numerical simulation

The study area considered here is a hypothetical heterogeneous irregular unconfined aquifer which is a modified version of Majumder and Eldho (2017) (See Fig. 6). Two rivers are flowing towards the north along the east and west boundaries which enforce the Dirichlet boundary condition. The head values vary linearly along the river boundaries. At the southernmost location, the head values are 760m and 720m for east and west boundaries respectively. The head value at the confluence, located at the north, is 710m. The

southern boundary is impervious or, no-flux boundary. A recharge of 6.5×10^{-3} m/d enters the aquifer through a recharge zone as shown in Fig. 6. The flow in the aquifer is assumed to be in steady-state. The total aquifer area of 29.13 km² is divided into four distinct zones of different hydrogeological parameters (See Fig. 6). The hydraulic conductivities of these zones are presented in Table 1. The hydraulic conductivity decreases from east to west direction with an exception of Zone 3 which has low hydraulic conductivity of $\kappa = 2$ m/d. The transport parameters i.e., longitudinal dispersivity and porosity values of these zones are presented in Table 1. The transverse dispersivity of a zone of the aquifer is assumed as 0.3 times the longitudinal dispersivity of the concerned zone. It may be noted that these values of the aquifer parameters are suitably assumed from existing literature (Ayvaz 2010; Li et al., 2020, 2021).

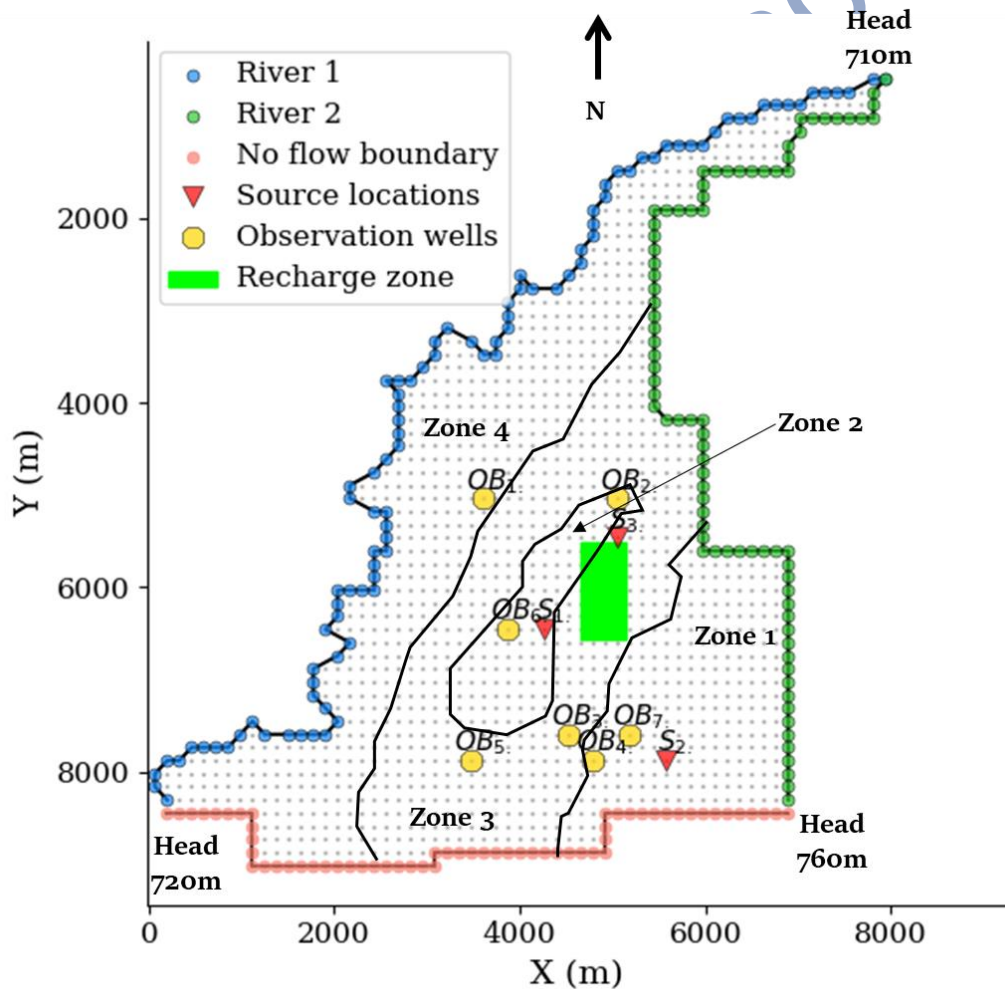


Fig. 6. Representation of study area for the case study

In this study, we assume three potential contamination sources (S₁, S₂ and S₃) located in the aquifer (See Fig. 6) which inject Tritium into the aquifer for 4 years in 4 stress periods each spanning 1 year. The

contaminant considered (Tritium) is reactive (half-life =12.32 years or, $\lambda=0.000153 \text{ d}^{-1}$). Hence, the total number of unknowns is 20 (3 possible source locations $\times$ 4 stress periods + 8 transport parameters). The concentrations are measured at seven observation points located at OB₁, OB₂, OB₃, OB₄, OB₅, OB₆ and OB₇ (See Fig. 6).

Table 1. Zonal parameters of the aquifer

Zones	Hydraulic conductivity (κ) (m/d)	Porosity (η_e)	Longitudinal Dispersivity (α_L) (m)
1	50	0.5	40
2	2	0.15	18
3	30	0.45	30
4	20	0.35	25

For the synthetic case study, numerically simulated concentrations are assumed as observed concentrations. The RPCM model is developed for the problem with 1585 nodes with a uniform nodal distance (d_c) of 131 m in x and y directions. As discussed, the shape functions are required to discretize the governing equations concerning the flow and transport process. These shape functions are estimated with MQ-RBFs with shape parameters q , α and local support domain size of 0.98, 5 and $3 \times d_c$ respectively. These parameters are adopted from the previous studies (Anshuman et al., 2019; Anshuman and Eldho, 2020; Singh et al., 2016) which provide detailed parameter sensitivity analysis. Further, RPCM simulations are also compared with widely used MODFLOW (Harbaugh, 2005) and MT3DMS (Zheng and Wang, 1999) in this study. The simulated head values are shown in Fig. 7 and Table 2. It is observed that the RPCM simulations match closely with that of MODFLOW at the observation points with a maximum percentage difference of 0.19.

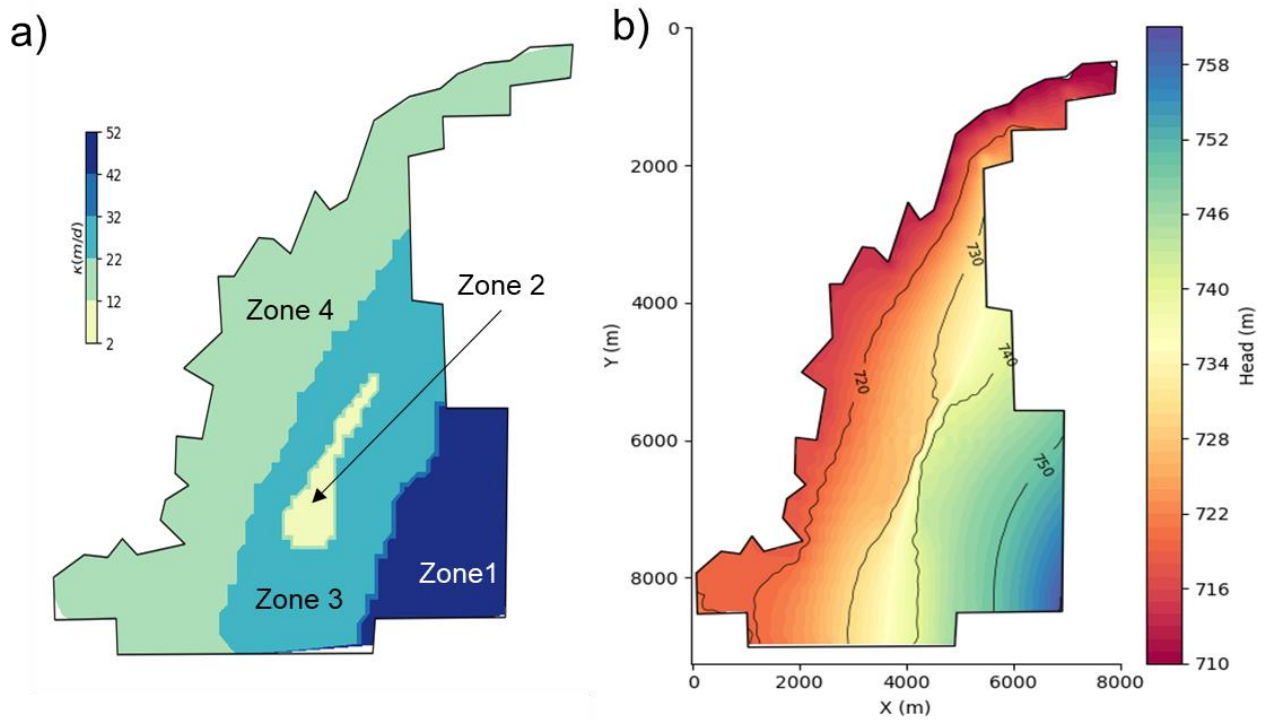


Fig. 7. a) Zones of the aquifer and corresponding hydraulic conductivities b) Head distribution in the aquifer as simulated by RPCM

Table 2. Comparison of RPCM and MODFLOW obtained heads at the observation points

Observation point	RPCM simulated heads(m)	MODFLOW simulated heads(m)	Percentage difference(%)
OB ₁	722.97	722.13	0.12
OB ₂	733.56	732.32	0.17
OB ₃	744.02	743.28	0.1
OB ₄	746.05	745.26	0.11
OB ₅	733.08	732.3	0.11
OB ₆	725.89	725.92	0
OB ₇	748.16	746.72	0.19

Considering the travel time required for contaminants to reach the observation points, the RPCM model is simulated for 15 years for the release rates given in Table 3. It may be noted that these rates are required to be estimated along with transport parameters in Table 2. Finer time step size increases the model stability at expense of computational time. Considering this, the time step size of 10 days is selected for simulation. The movement of contamination plumes in the aquifer throughout the simulation period are presented in Fig. 8. The RPCM and MT3DMS simulated breakthrough curves (BTC) at the observation points are presented in Fig. 9. The error indicators mean absolute error (MAE (ppm)), root mean squared

error (RMSE (ppm)) and normalized RMSE are computed for the BTC obtained at observation locations (See Table 4). The normalization is done with respect to the maximum concentration obtained by the MT3DMS model at the observation wells. At some locations, the RPCM simulated BTCs are observed to rise earlier/later compared to that of MT3DMS as shown in Fig. 9. Such minor differences are generally observed when two different numerical models are compared especially in the case of a heterogeneous aquifer with irregular boundaries (Pathania et al., 2019; Anshuman and Eldho, 2020). Apart from that, RPCM model simulations match very well with the results of MT3DMS with maximum values of MAE, RMSE and normalized RMSE being 1.098 ppm, 1.421 ppm and 0.152 respectively. The computational time required for the simulation is 34.39 seconds using an Intel i7 10th Gen system having 16 GB RAM and 2.2 GHz which is aimed to be reduced using the EAS-LSTM surrogate model.

Table 3. Release rate at the source locations at different stress periods

Stress period	Concentration release (ppm)		
	S ₁	S ₂	S ₃
1	5	25	6
2	10	18	14
3	28	7	20
4	5	10	25

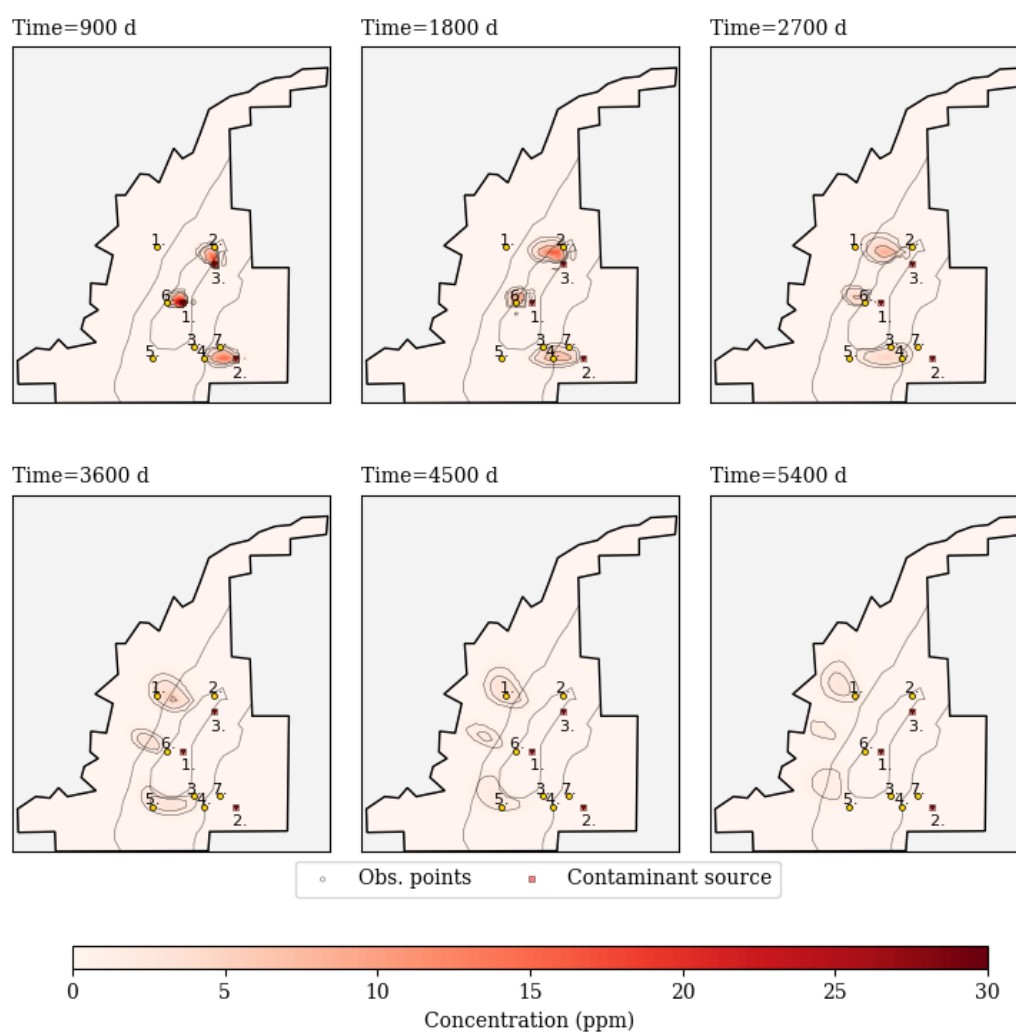


Fig. 8 Movement of contamination plumes in the aquifer through different time instants

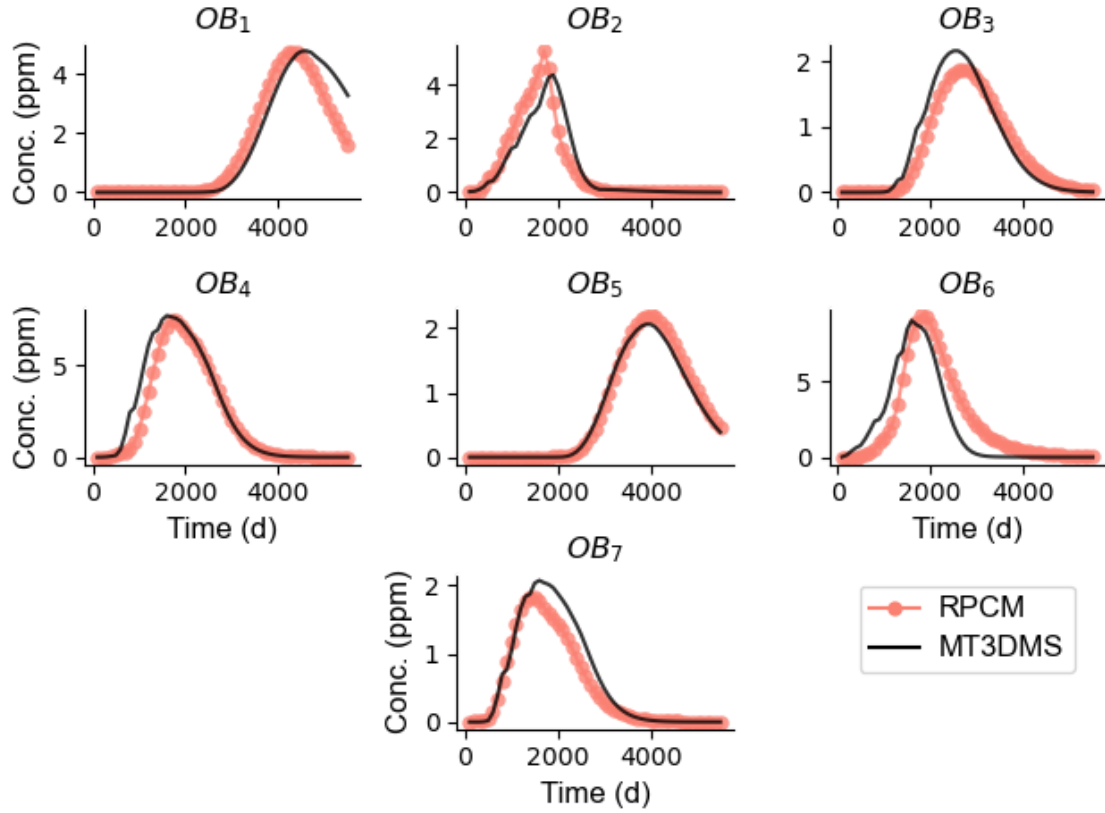


Fig. 9. Concentration variation with time at different observation points simulated by RPCM in comparison with MT3DMS

Table 4. Error metrics for concentration profiles obtained by RPCM models with respect to MT3DMS simulations at observation points

Observation point	MAE (ppm)	RMSE (ppm)	Normalised RMSE
OB1	0.399	0.629	0.067
OB2	0.350	0.609	0.065
OB3	0.150	0.208	0.022
OB4	0.364	0.786	0.084
OB5	0.063	0.091	0.010
OB6	1.098	1.421	0.152
OB7	0.134	0.220	0.024

3.2 Objective function

As mentioned in section 3.1, the release rates at the possible source locations and transport parameters i.e., longitudinal dispersity and porosity are required to be estimated. For this inverse modelling problem, the error between simulated and observed concentrations at the observation points for contamination release

history is minimized. The mathematical formulation of the objective function is given by (Guneshwor et al., 2018)

$$\min \frac{1}{n_{obs}} \sum_{i=1}^{n_{obs}} \sum_{j=1}^{n_t} (c_{norm_{ij}}^{obs} - c_{norm_{ij}}^{sim})^2 \quad (40)$$

$$\text{Subject to } \begin{cases} Q_L < Q_N < Q_U, N \in [1,3] \\ \alpha_L^L < \alpha_L^n < \alpha_L^U, n \in [1,4] \\ \eta_e^L < \eta_e^n < \eta_e^U, n \in [1,4] \\ c^{sim} = f(Q, \alpha_L, \eta_e) \end{cases} \quad (41)$$

Where c_{norm}^{obs} and c_{norm}^{sim} are normalized observed concentrations and surrogate model simulated concentrations at the observation points respectively. The terms n_{obs} and n_t represent the number of observation wells and the number of stress periods respectively. The normalized concentrations at the observation points, c_{norm}^{sim} , is a function of contamination strength Q at N source locations and aquifer parameters (α_L and η_e) at n zones as shown in Eq. (41). The superscripts L and U in the constraint terms i.e., Eq. (41) denote the lower and upper bounds of the variables used in this study which are also presented in Table 5.

Table 5. Upper and lower bounds of the variables to be estimated by the optimization model

Variable	Lower bound (L)	Upper bound(U)
Release rates at source locations (Q) (ppm)	0	40
Longitudinal dispersivity (α_L) (m)	10	50
Porosity (η_e)	0.1	0.6

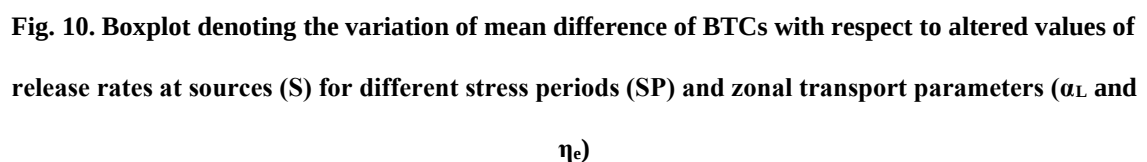
3.2.1 Erroneous concentrations

It is evident from the objective function i.e., Eq. (40) that the performance of the model significantly depends on the accuracy of the observed concentration data at the observation wells. However, in practice, the measurement of the observed concentrations at the observation wells is susceptible to human and instrumental errors (Yeh et al., 2007). Considering this, the robustness of the proposed model is analyzed using perturbed observation data by using a random normal deviate given by

$$c'_{obs} = c_{obs} \times [1 + \mathcal{N}(0, \sigma)] \quad (42)$$

Here $\mathcal{N}(0, \sigma)$ denotes the random normal deviate of mean 0 and standard deviation σ . In this study, σ is taken as 0.1. The function *randn* from Numpy (Oliphant 2006) is used for generating random numbers. The observed concentration in the objective function in Eq. (40) is replaced with perturbed observed concentrations (c'_{obs}). The mean deviations from the c_{obs} is 0.016 ppm.

In SSO, the optimization model searches for unknowns i.e., contaminant release histories and transport parameters by minimizing the objective function explained in Eq. (40) which depends on the difference between the simulated and observed BTCs at the observation points. Here the impact of the unknown variables on this difference is investigated. Each variable is changed by -50% to +50% of its actual value, keeping the other variables unchanged, as denoted in Tables 1 and 3 with increments of 10%. For each analysis, the variables other than the concerned variable are fixed as in Tables 1 and 3. The mean difference in normalized observed and simulated concentrations for the altered variable is plotted in Fig. 10. To give equal weightage to all observation points, the observed and simulated BTCs are normalized. It is observed that the mean difference is most sensitive to the porosity of zone 1 and zone 3. The porosity of the medium affects the velocity and the occurrence of peak concentration. As seen in Fig. 8, the contaminant released from S_2 located in zone 1 impacts 4 out of a total of 7 observation points in the aquifer. The high sensitivity of the mean difference to the porosity values corresponding to these zones is explained by the presence of these observation points in zone 1 and zone 3. Similarly, the mean difference is more sensitive to the dispersivity values of these zones compared to other zones.



3.4 Surrogate simulation optimization

The steps involved in the development of surrogate simulation optimization model are given as follows:

1. **Data pre-processing:** The input data for EAS-LSTM training comprises dynamic or, sequential data i.e., contamination strengths at source locations and static data i.e., transport parameters (α_L and η_e). For each set of input data, RPCM model simulated BTCs at the observation points are used as output. Uniform random sampling is used to generate input data for the given sample size and corresponding simulated concentrations are estimated by RPCM. These data are normalized between respective maximum and minimum values to facilitate the training of the surrogate model. In this study, a total of 2500 samples are generated out of which 2000 samples are used for model training and the accuracy of the trained model is tested on 500 samples. It may be noted that the neural network during the training process is blind to the samples used for the testing. The time required for the generation of the sample is 2.86 hours. The sequential and static input data are arranged in $n_{sample} \times n_{sp} \times n_s$ and $n_{sample} \times n_T$ forms respectively. Here n_{sample} , n_{sp} , n_s and n_T denote numbers of samples, stress periods, potential sources and transport parameters respectively. Similarly, the output data is arranged in $n_{sample} \times n_t \times n_o$. Here n_t and n_o represent numbers of time steps in BTCs and observation points respectively. It is to be noted that the prior knowledge of the groundwater flow system and BTCs at the observation points assists in determining the potential sources of the contamination.
2. **EAS-LSTM model architecture:** EAS-LSTM model is developed in PyTorch (Paszke et al., 2019). The input data is fed into the encoder EALSTM and encoded into hidden states C_t and H_t . For decoder LSTM, a zero matrix of shape $n_s \times 1 \times n_o$ is provided as *teacher forcing* for the first timestep. The encoded states from encoder EALSTM i.e., C_t and H_t are set as initial states for decoder LSTM. For the subsequent timesteps, the output from the previous LSTM cell is provided as input. The output from the decoder LSTM is set as input to a dense layer with linear activation.
3. **Training:** The EAS-LSTM model is trained using the ADaptive Moment estimation (ADAM) algorithm (Kingma and Ba, 2014). The trainable model parameters (weights and biases) are optimized using the gradient descent technique by minimizing the mean squared difference between the training data and predicted data. The model is trained for 2500 epochs using a batch size of 200. The trainable model parameters i.e., weights and biases in Eq. (21-24) are saved at the

end of each epoch where the testing MSE betters its previous value. The model parameters corresponding to the least values of testing MSE are used for prediction.

4. **Evaluation:** Once the EAS-LSTM model is trained, it can be used for prediction. The sequential and static input data for the trained model is required to be in the same format as step 1. The encoder EALSTM converts the input data into an encoded state which is further decoded by the decoder LSTM for computation of output sequence. As described in step 2, the decoder LSTM necessitates a zero tensor as teacher forcing for the first timestep. The model performance is evaluated for the training and testing dataset.
5. **Optimization:** A SSO model is developed using EAS-LSTM, MVO and the objective function. The MVO algorithm is utilized to estimate the unknown variables i.e., contamination release rates and transport parameters by minimizing the objective function described in Eq. (40) which uses EAS-LSTM model simulations. The population for the variables is initiated within the upper and lower bounds. It is to be noted that the inputs to the EAS-LSTM model are required to be normalized using *max-min* normalization as shown in Eqs (43-45).

$$Q_{norm}^i = \frac{Q^i - \min(Q)}{\max(Q) - \min(Q)} \quad (43)$$

$$T_{norm}^i = \frac{T^i - \min(T)}{\max(T) - \min(T)} \quad (44)$$

$$C_{norm}^i = \frac{C^i - \min(C)}{\max(C) - \min(C)} \quad (45)$$

Here Q and T imply the dynamic/ sequential and static input variables respectively. C denotes the output variable. It is to be noted that the maximum and minimum values of Q, T and C are computed for the entire sample size. The predictions of EAS-LSTM, which are normalized values, are scaled up to provide the required output sequence. These predictions are used for computing the objective function which is to be minimized iteratively. The best solution is updated at each iteration.

The methodology used for solving the source identification and parameter estimation problem using EAS-LSTM and MVO is presented in Fig. 11.

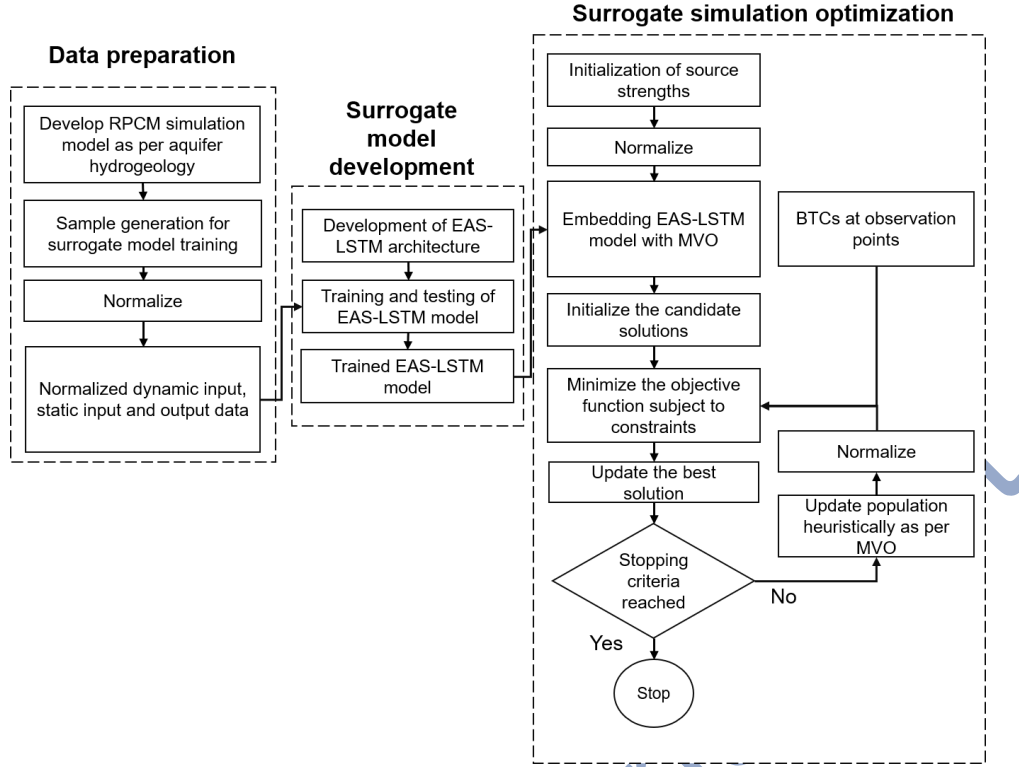


Fig. 11. Modelling framework for solving the source identification and parameter estimation problem using EAS-LSTM and MVO

3.5 Performance evaluation

The performance of an SSO model depends on the accuracy of the surrogate model to approximate the simulation model and the ability of the optimization algorithm to find the global optimum. The following error metrics are used to evaluate the performance of the SSO model.

$$R^2 = 1 - \frac{\sum_{i=1}^{n_{obs}} \sum_{j=1}^{n_t} (C_{ij}^{obs} - C_{ij}^{sim})^2}{\sum_{i=1}^{n_{obs}} \sum_{j=1}^{n_t} (C_{ij}^{obs} - \overline{C_{ij}^{sim}})^2} \quad (46)$$

$$RMSS = \sqrt{\frac{\sum_{i=1}^{n_{source}} \sum_{j=1}^{n_t} (Q_{ij}^{act} - Q_{ij}^{pred})^2}{n_{source}}} \quad (47)$$

$$RMST = \sqrt{\frac{\sum_{i=1}^{n_T} (T_i^{act} - T_i^{pred})^2}{n_T}} \quad (48)$$

$$RMSO = \sqrt{\frac{\sum_{i=1}^{n_{obs}} \sum_{j=1}^{n_t} (C_{ij}^{obs} - C_{ij}^{sim})^2}{n_{obs}}} \quad (49)$$

Here Q and T denote the contamination strengths at the source locations and transport parameters respectively. The superscripts *act* and *pred* correspond to actual and predicted values respectively. The terms n_{source} , n_T , n_{obs} , n_{sp} and n_t denote the number of potential source locations, transport parameters, observation points, spatial locations and transport parameters respectively.

observation well, stress periods and timesteps respectively. In this study, the performance of EAS-LSTM is compared with that of Kriging and SVR. The MVO, GWO and PSO are used as optimization algorithms. The accuracy of the SSO models is signified by the proximity of R^2 to 1 and low values of RMSS, RMST and RMSO.

4 Model applications

4.1 Predictive accuracy of surrogate models

The surrogate models EAS-LSTM, Kriging and SVR are used for the prediction of BTCs at the observation wells for given contaminant release histories at source locations and transport parameters. A trained surrogate model must approximate the simulation model accurately to be effectively used in combination with optimization to estimate unknown variables. The training and testing data set used for all the three surrogate models are the same as presented in section 3.4. To test the generalization capability of the surrogate models the K -fold cross-validation technique (Refaeilzadeh et al., 2009) is used where the value of K is set to 5. The mean convergence curve of the EAS-LSTM model using different folds/sets of training samples is presented in Fig. 12. The minimum value of mean squared error (MSE) obtained for the training is $1.7 \times 10^{-6} \text{ ppm}^2$. The mean values of different error indicators i.e., MAE (ppm), RMSE (ppm) and R^2 for training and testing datasets are compared in Fig. 13 for all surrogate models. It is observed that the proposed EAS-LSTM model provides lower values of MAE and RMSE and a higher value of R^2 in comparison to Kriging and SVR models. It may be noted that the standard deviation values obtained by the EAS-LSTM models using different combinations of training and testing data are low as shown in Fig. 13.

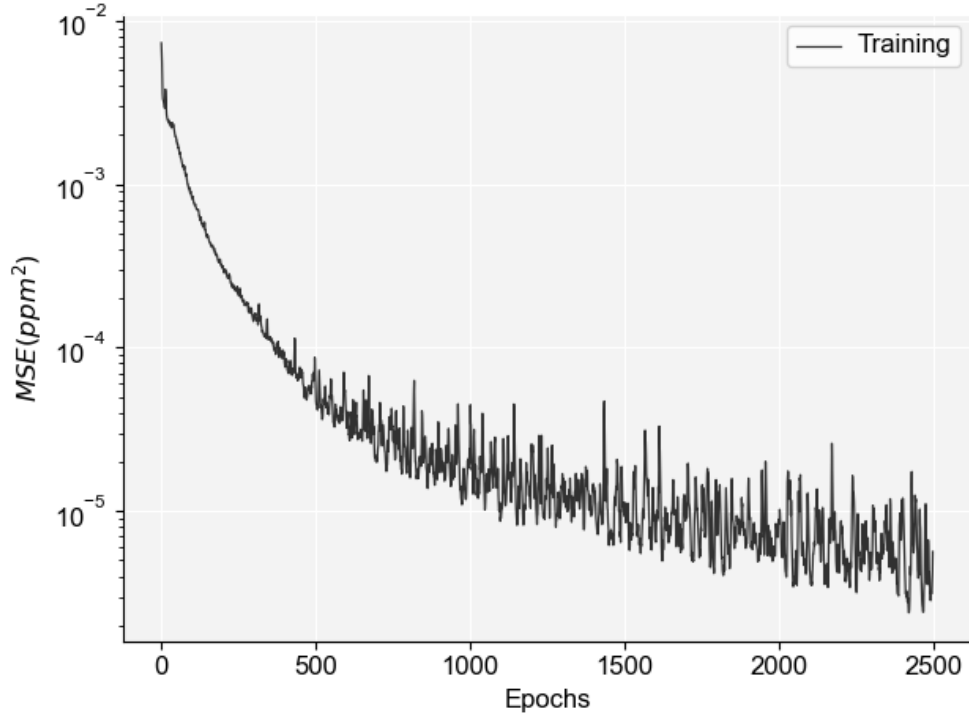


Fig. 12. Mean convergence curve for the training of EAS-LSTM models using 5 folds of training

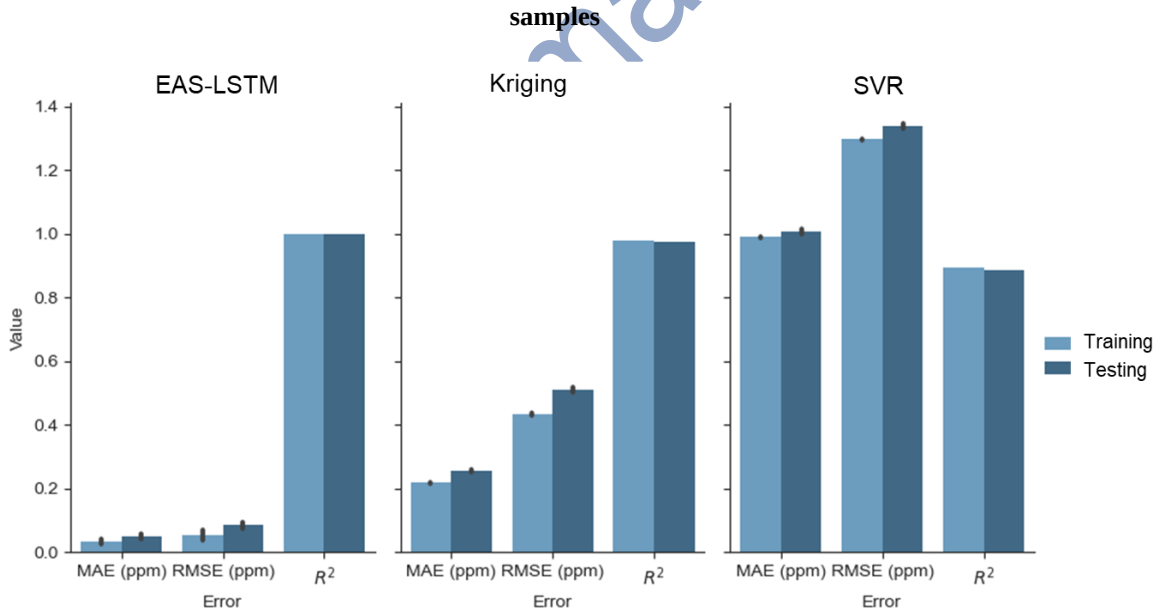


Fig.13. Bar plots representing the mean values of error indicators and standard deviations (as error bars) for training and testing data for different surrogate models

The BTCs at different observation wells simulated by the surrogate models for the current study are presented in Fig. 14 using the input variables in Table 1 and Table 3. The EAS-LSTM model is observed to reconstruct the actual BTCs more accurately compared to Kriging and SVR. The corresponding MSE and R^2 values are presented in Table 6 which shows that the EAS-LSTM model predictions are most accurate. It

may be noted that the MSE and R^2 are computed using concatenated BTCs. The time required per simulation in EAS-LSTM is 0.031s which is higher than that of Kriging. However, it is 1066.5 times faster than the simulation model which takes 33.06s. The training of EAS-LSTM is done in NVIDIA GeForce 1660 Ti GPU. The training time is significantly higher than that of Kriging and SVR models which is generally the case in LSTMs based deep neural networks (Li et al., 2021).

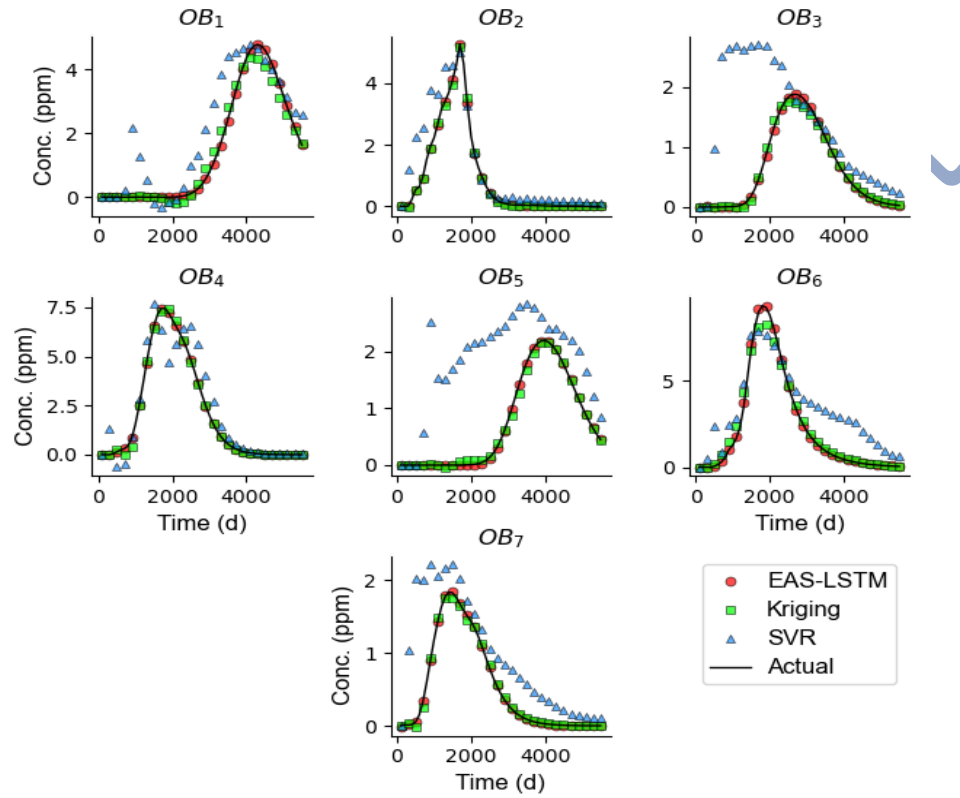


Fig.14. Comparison of breakthrough curves obtained at different observation wells using different surrogate models

Table 6. Comparison of performance for different surrogate models

	Training duration(s)	Simulation time (s)	MSE (ppm ²)	R ²
EAS-LSTM	2247	0.031	0.0033	0.9995
Kriging	9.66	0.006	0.048	0.9930
SVR	2.82	0.046	1.302	0.885

4.2 Inverse modelling results

4.2.1 Scenario 1

In this inverse modelling problem, the release histories at source locations and zonal transport parameters are required to be estimated using the BTCs at the observation points. For the current scenario, no error in the contaminant measurement is assumed i.e., model-simulated concentrations at the observation points (see Fig.9) are used for the formulation of the objective function in Eq. (40). The simulations from the surrogate models i.e., EAS-LSTM, Kriging and SVR, which drastically reduce the computational time, are used as c^{sim} and used in the objective function. The unknowns are estimated by three optimization models MVO, GWO and PSO using population size and maximum iteration number of 100 and 3000 respectively which indicates that the surrogate models are called 300,000 times by the optimization algorithms. The upper and lower bounds of the unknowns are set as per Table 5. The evolution of objective function with iteration number for the SSO models is presented in Fig. 15. As seen in Table 7, the minimum value of the objective function obtained is least for EAS-LSTM and MVO compared to its counterparts.

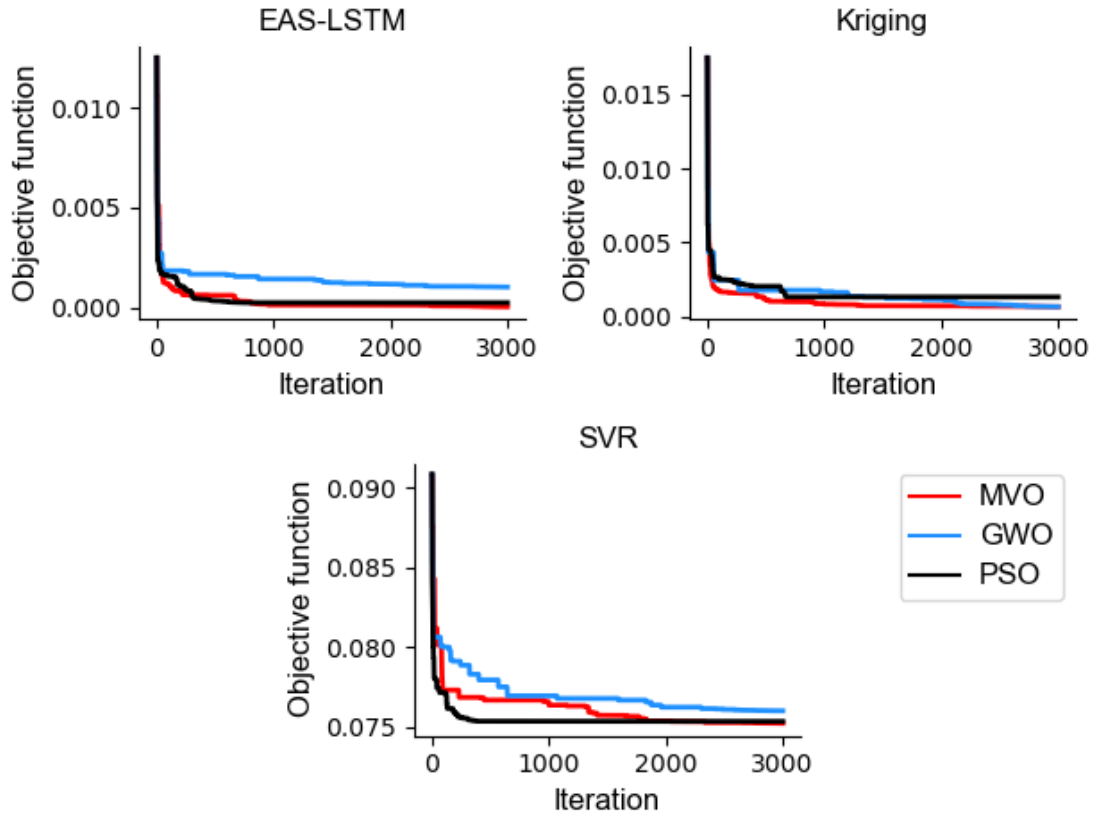


Fig. 15. Convergence curves of different SSO models

Table 7. Minimum values of the objective function (ppm²) obtained by different SSO models

	MVO	GWO	PSO
EAS-LSTM	0.000045	0.001030	0.000249
Kriging	0.000658	0.000663	0.001317
SVR	0.075198	0.075984	0.075320

The 20 estimated unknown variables are presented in Fig. 16. The EAS-LSTM-MVO model is observed to give accurate results compared to other counterparts. The error metrics RMSS and RMST obtained are the least for EAS-LSTM-MVO as seen in Table 8 which denotes that the release histories and transport parameters are accurately estimated. Further, the low value of RMSO indicates that the EAS-LSTM can reconstruct the BTCs at the observation points more precisely using MVO estimated variables. Similarly, the performance of MVO and GWO are similar and are better than that of PSO using Kriging. Nonetheless, the reconstructed BTCs using Kriging and SVR are not as accurate as EAS-LSTM (See Fig. 14 and Table 8) resulting in an imprecise estimation of the unknowns.

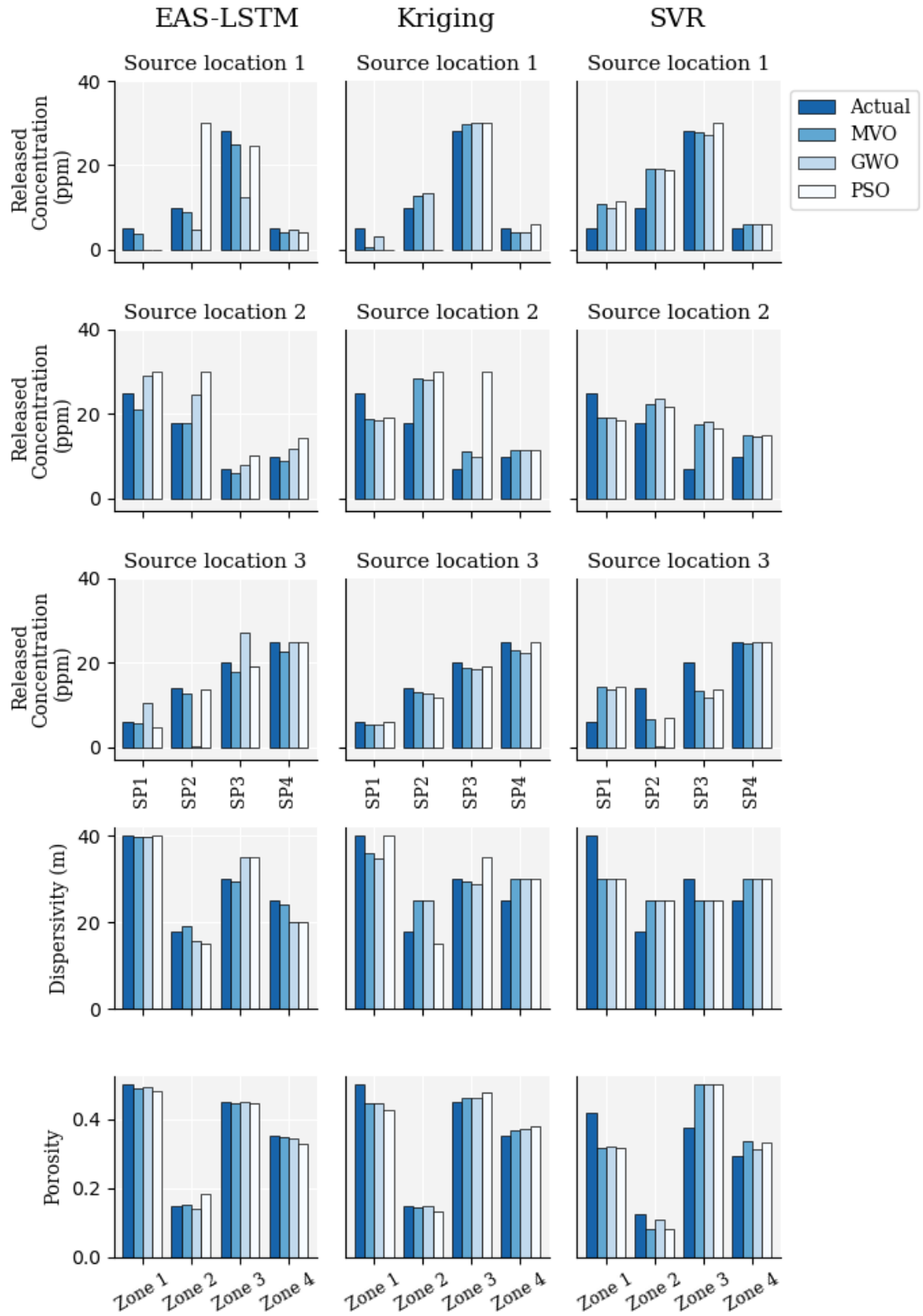


Fig. 16. Comparison of estimated variables using different SSO models

Table 8. Estimated error metrics of different SSO models

Surrogate models	Optimization algorithms	R ²	RMSS	RMST	RMSO
EAS-LSTM	MVO	0.9954	1.9102	0.5112	0.2589
	GWO	0.8878	7.2188	2.6187	0.8353
	PSO	0.8941	7.2860	2.7157	0.6428
Kriging	MVO	0.9471	4.1070	3.3783	0.4280
	GWO	0.9459	3.9125	3.6178	0.3893
	PSO	0.8675	8.3939	2.7159	0.8428
SVR	MVO	0.8657	6.3070	4.9880	1.2462
	GWO	0.8300	7.3094	4.9879	1.2449
	PSO	0.8700	6.1964	4.9880	1.2305

4.2.2 Scenario 2

Given the vulnerability of the observed concentrations to human and instrumental errors, perturbed observed concentrations c'_{obs} as mentioned in section 3.2.1 are used in the objective function in Eq. (40). These concentrations at different observation points are presented in Fig. 17. The SSO models discussed in section 4.2.1 for all noise levels are run to estimate the unknowns. The population size and number of iterations are the same as described in section 4.2.1 for all optimization models. The results obtained are presented in Fig. 18 and Table 9. It is observed that the EAS-LSTM-MVO model provides low values of RMST and RMSO. However, the RMSS and RMSO values obtained by EAS-LSTM-GWO is lower compared to EAS-LSTM-MVO. Interestingly, Kriging-MVO and Kriging-GWO models give slightly better results than that of the EAS-LSTM-MVO model in terms of R² which denotes better estimation of unknowns. However, as per the mathematical formulation of the objective function, the EAS-LSTM-MVO converged to a lesser value as observed in Table 10. On the contrary, the error metrics concerning the SVR surrogate model denotes its poor performance to obtain the unknowns. The reason can be attributed to the low accuracy of the trained SVR surrogate model to reconstruct the unperturbed BTCs at observation wells as observed in section 4.1.

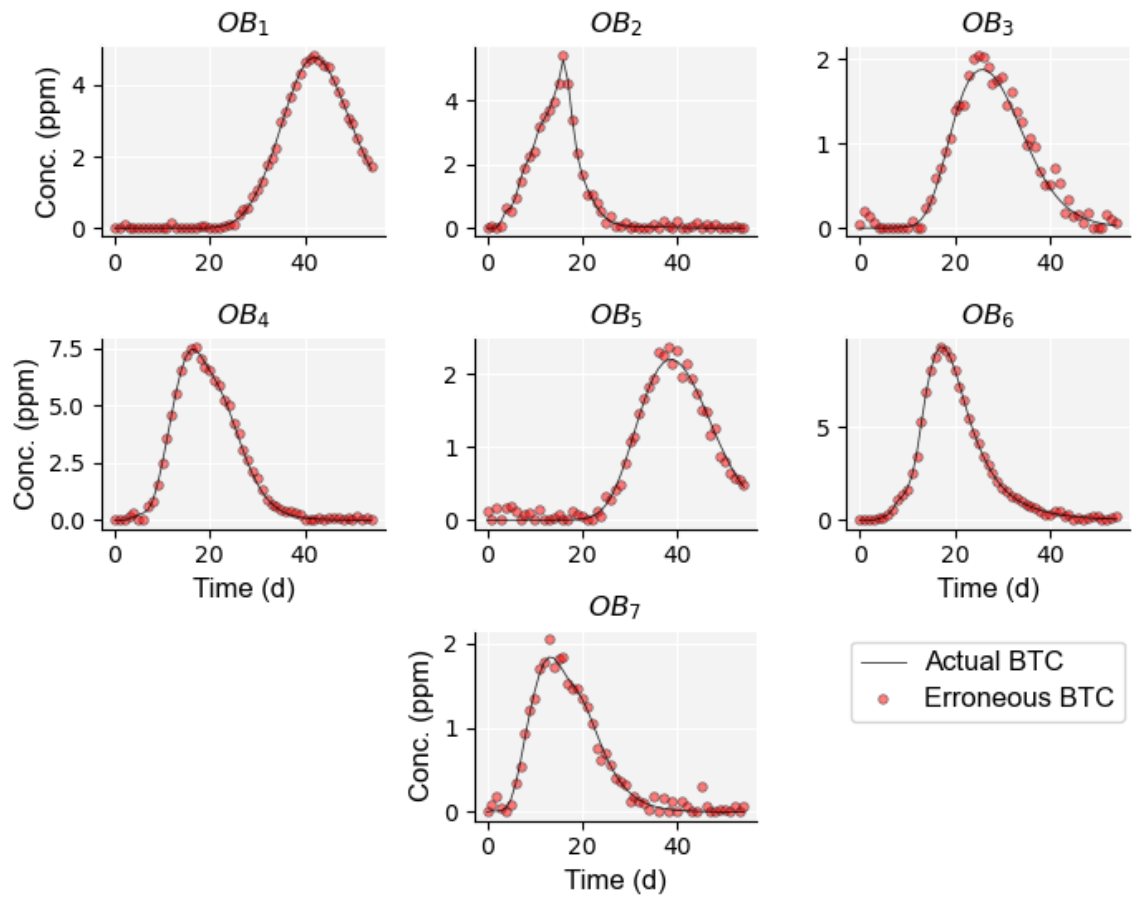


Fig. 17. Representation of erroneous BTCs at observation points

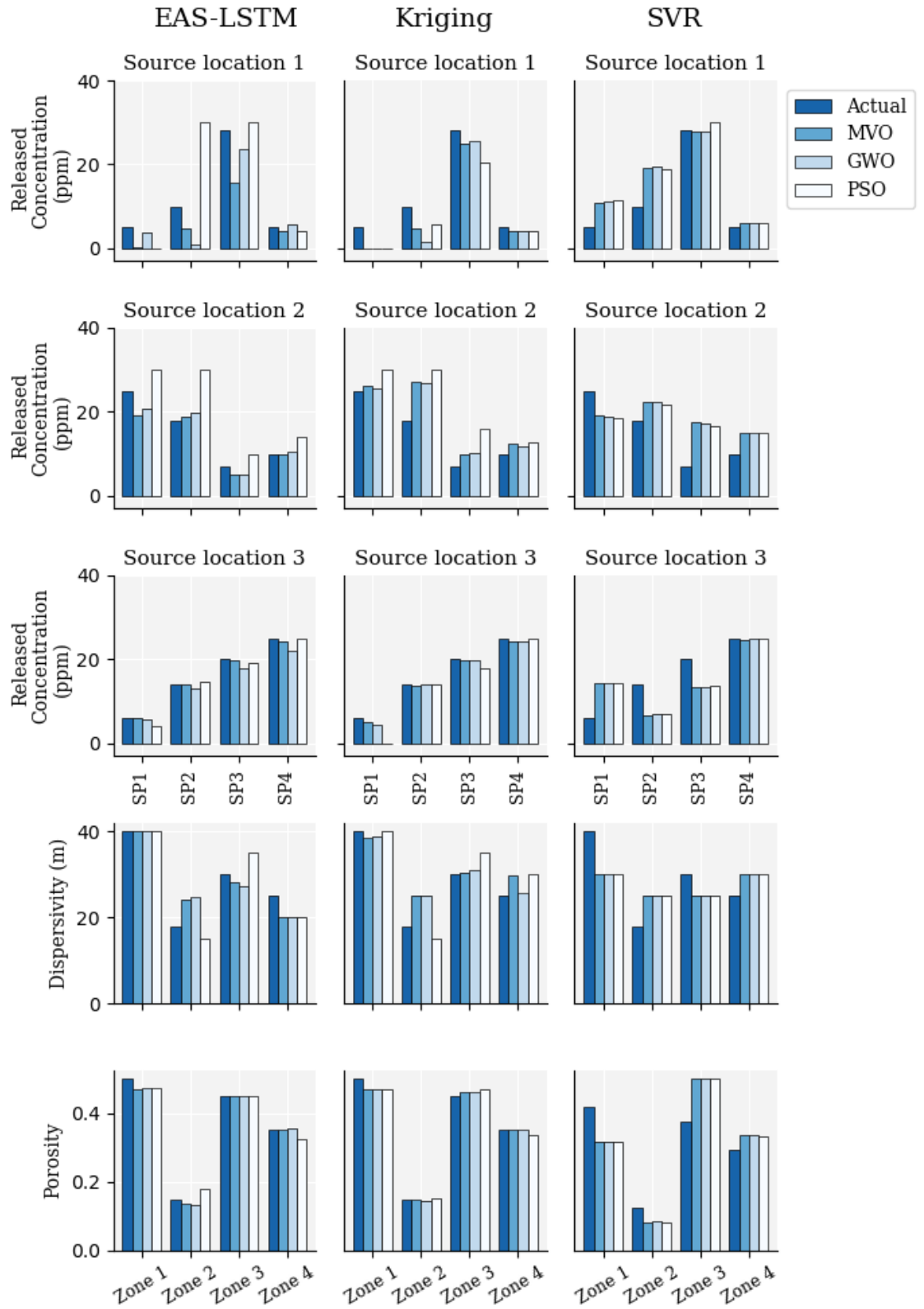


Fig. 18. Comparison of estimated variables using different SSO models for perturbed observed concentrations

Table 9. Estimated error metrics of different SSO models for perturbed observed concentrations

Surrogate model	Optimization algorithm	R ²	RMSS	RMST	RMSO
EAS-LSTM	MVO	0.95	4.52	2.84	0.72
	GWO	0.96	3.42	3.12	0.47
	PSO	0.90	7.23	2.72	0.74
Kriging	MVO	0.96	3.65	3.07	0.61
	GWO	0.96	4.07	2.54	0.63
	PSO	0.93	5.76	2.72	0.81
SVR	MVO	0.87	6.31	4.99	1.24
	GWO	0.87	6.33	4.99	1.24
	PSO	0.87	6.20	4.99	1.23

Table 10. Comparison of minimum values of objective functions obtained

Methods		Minimum value of the objective function
EAS-LSTM	MVO	0.0011
	GWO	0.0012
	PSO	0.0014
Kriging	MVO	0.0016
	GWO	0.0018
	PSO	0.0022
SVR	MVO	0.0241
	GWO	0.0242
	PSO	0.0236

5 Discussion

The current study proposes a novel DNN named EAS-LSTM for the estimation of contaminant release histories at potential sources and zonal transport parameters in the aquifer for given observed contamination data. The DNN uses an EALSTM (Kratzert et al., 2019) for taking sequential and static components of the input and transforming them into an encoded state. The timestep wise outputs are generated by a decoder LSTM recursively. The developed model is applied to a heterogeneous aquifer with 4 distinct zones having known hydraulic conductivities. It is required to estimate the release intensities of Tritium from 3 prospective source locations and zonal transport parameters i.e., longitudinal dispersivity and effective porosities for observed concentrations at 7 observation points scattered throughout the aquifer. The data required for the training of the EAS-LSTM is generated using the meshfree RPCM model. The MAE, RMSE and R² values obtained using K-fold cross-validation indicate that the EAS-LSTM performs better compared to Kriging and SVR. Consequently, the EAS-LSTM model reproduces the BTCs at 7 observation wells compared to the other two surrogate models. The training time requirement of EAS-LSTM is much higher

than that of Kriging and SVR which is expected as LSTMs are more computationally intensive. However, the prediction time is only 0.031s which is 1066.5 times lower than the simulation model i.e., 33.06s. This drastic reduction in computational time allows for higher population size and iteration numbers in inverse modelling.

For inverse modelling, three stochastic population-based optimization algorithms having unique search space exploration strategies i.e., MVO, GWO and PSO were utilized using surrogate model simulations. The PSO implements the movement of the bird flock by updating the position of the candidate solutions based on the best positions of the individual candidates and overall swarm population as shown in Eqs. (38) and (39). The GWO implements the hunting behaviour of the grey wolves and the position of candidate solutions are updated by encircling the prey principle as per Eqs. (36) and (37). In contrast to PSO and GWO, the MVO algorithm changes the candidate solutions through two steps. It is accomplished through roulette wheel selection and local exploitation using WEP and TDR parameters (see Eqs. (32)-(35)). The roulette wheel selection is accomplished at each iteration to facilitate the exchange of information between the best and other candidate solutions. The WEP parameter ensures that the MVO algorithm makes a smooth transition from the dominance of the exploration phase to the exploitation phase with an increase in the number of iterations. The SSO models are applied to two scenarios. In the first scenario, the BTCs at the observation points are considered error-free while in the second, a random noise is added deliberately to the evaluation of SSO model performances. It is observed that the combination EAS-LSTM and MVO provides better results in comparison to other surrogate models in estimating the unknown release rates and transport parameters. Moreover, the BTCs obtained EAS-LSTM-MVO estimated variables also closely match with observed concentrations yielding less value of RMSO. In scenario 2, the performances of Kriging-GWO and Kriging-PSO models are better in terms of R^2 . Nonetheless, the minimum value obtained by EAS-LSTM-MVO is lower than that of the aforementioned SSO models which denote that the minimization of the objective function is handled better by MVO. Some previous studies have also indicated the superiority of the MVO against GWO and PSO (Benmessahel et al., 2018; Kumar et al., 2019). Considering the time required for sample generation (2.86 hours), training (0.62 hours) and optimization (2.6 hours), the computational time required for the EAS-LSTM-MVO model is 6.08 hours. On the other hand, if the optimization model is embedded with the simulation model, the computational time required would approximately be 2755 hours for the same population size and iteration number in the optimization algorithm. Moreover, once the EAS-LSTM model is trained, it can be used repetitively for different analyses

which means that the cost required for optimization will only be considered. Although the proposed SSO framework has demonstrated significant advantages compared to other SSO models discussed in the study, it is noteworthy that the time required for training is more in EAS-LSTM and it requires Graphical Processing Units (GPU) as training is prohibitively expensive for a large sample size and hidden neurons without GPUs. Furthermore, in the current study, the lengths of the input and output sequences are unequal and the output sequence (55 steps) is over 13 times lengthier compared to the input sequence (4 steps). For simpler problems having shorter sequence lengths, shallow machine learning algorithms may be more suitable (Wunsch et al., 2021).

6 Conclusions

The source identification and parameter estimation are conventionally done using the S/O approach. However, as S/O models are computationally expensive, a modelling framework using a surrogate model named EAS-LSTM is proposed in this study which uses EALSTM as an encoder and LSTM as a decoder. The performance comparison of Kriging and SVR models suggests that the proposed surrogate model can approximate the simulation model more accurately although the time required for training is higher in the case of the EAS-LSTM model. Amongst the SSO models developed for estimation of unknowns, the combination of EAS-LSTM and MVO was found to be more effective to estimate the unknown variables i.e., release rates at source locations and transport parameters compared to other SSO models. Moreover, the EAS-LSTM is over 1000 times faster than the simulation model. Owing to the accuracy and computational efficiency of the surrogate model, the EAS-LSTM model can be efficiently used for source identification and parameter estimation of large-scale field problems.

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CRediT author statement

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References

- Al-Fugara, A. K., Ahmadlou, M., Al-Shabeeb, A. R., AlAyyash, S., Al-Amoush, H., & Al-Adamat, R. (2020). Spatial mapping of groundwater springs potentiality using grid search-based and genetic algorithm-based support vector regression. *Geocarto International*, 1-20.
- Anshuman, A., Eldho, T. I., & Singh, L. G. (2019). Simulation of reactive transport in porous media using radial point collocation method. *Engineering Analysis with Boundary Elements*. 104, 8-25.
- Anshuman, A., & Eldho, T. I. (2020). Meshfree radial point collocation-based coupled flow and transport model for simulation of multispecies linked first order reactions. *Journal of Contaminant Hydrology*. 229, 103582.
- Aral, M. M., Guan, J., & Maslia, M. L. (2001). Identification of contaminant source location and release history in aquifers. *Journal of Hydrologic Engineering*. 6(3), 225-234.
- Ayvaz, M. T. (2010). A linked simulation–optimization model for solving the unknown groundwater pollution source identification problems. *Journal of Contaminant Hydrology*. 117(1-4), 46-59.
- Batu, V. (2005). Applied flow and solute transport modeling in aquifers: fundamental principles and analytical and numerical methods. CRC Press.
- Bear, J., & Cheng, A. H. D. (2010). Modeling groundwater flow and contaminant transport (Vol. 23). Springer Science & Business Media.
- Bear, J., & Verruijt, A. (2012). Modeling groundwater flow and pollution (Vol. 2). Springer Science & Business Media.
- Benmessahel, I., Xie, K., & Chellal, M. (2018). A new evolutionary neural networks based on intrusion detection systems using multiverse optimization. *Applied Intelligence*, 48(8), 2315-2327.
- Dhar, A., & Datta, B. (2009). Global optimal design of ground water monitoring network using embedded kriging. *Groundwater*, 47(6), 806-815.
- Eardley, D. M. (1974). Death of white holes in the early universe. *Physical Review Letters*. 33(7), 442.

- Gao, S., Huang, Y., Zhang, S., Han, J., Wang, G., Zhang, M., & Lin, Q. (2020). Short-term runoff prediction with GRU and LSTM networks without requiring time step optimization during sample generation. *Journal of Hydrology*, 125188.
- Guneshwor, L., Eldho, T. I., & Kumar, A. V. (2018). Identification of groundwater contamination sources using meshfree RPCM simulation and particle swarm optimization. *Water Resource Management*, 32(4), 1517-1538.
- Gurarslan, G., & Karahan, H. (2015). Solving inverse problems of groundwater-pollution-source identification using a differential evolution algorithm. *Hydrogeology Journal*. 23(6), 1109-1119.
- Harbaugh, A. W. (2005). *MODFLOW-2005, the US Geological Survey modular ground-water model: the ground-water flow process* (pp. 6-A16). Reston, VA: US Department of the Interior, US Geological Survey.
- Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735-1780.
- Kansa, E. J. (1990 a). Multiquadrics—A scattered data approximation scheme with applications to computational fluid-dynamics—I surface approximations and partial derivative estimates. *Computers & Mathematics with Applications*. 19(8-9), 127-145.
- Kansa, E. J. (1990 b). Multiquadrics—A scattered data approximation scheme with applications to computational fluid-dynamics—II solutions to parabolic, hyperbolic and elliptic partial differential equations. *Computers & Mathematics with Applications*. 19(8-9), 147-161.
- Kao, I. F., Zhou, Y., Chang, L. C., & Chang, F. J. (2020). Exploring a Long Short-Term Memory based Encoder-Decoder framework for multi-step-ahead flood forecasting. *Journal of Hydrology*, 124631.
- Kennedy, J., & Eberhart, R. (1995, November). Particle swarm optimization. In Proceedings of ICNN'95-International Conference on Neural Networks (Vol. 4, pp. 1942-1948). IEEE.
- Kingma, D. P., & Ba, J. (2014). Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*.
- Kratzert, F., Klotz, D., Shalev, G., Klambauer, G., Hochreiter, S., & Nearing, G. (2019). Towards learning universal, regional, and local hydrological behaviors via machine learning applied to large-sample datasets. *Hydrology and Earth System Sciences*, 23(12), 5089-5110.
- Kumar, M. B. H., Balasubramanian, S., Padmanaban, S., & Holm-Nielsen, J. B. (2019). Wind energy potential assessment by weibull parameter estimation using multiverse optimization method: A case study of Tirumala region in India. *Energies*, 12(11), 2158.
- Li, J., Lu, W., Wang, H., Fan, Y., & Chang, Z. (2020). Groundwater contamination source identification based on a hybrid particle swarm optimization-extreme learning machine. *Journal of Hydrology*, 584, 124657.
- Li, J., Lu, W., & Jiannan, L. (2021). Groundwater contamination sources identification based on the Long-Short Term Memory network. *Journal of Hydrology*, 126670.
- Liang, Z., Zou, R., Chen, X., Ren, T., Su, H., & Liu, Y. (2020). Simulate the forecast capacity of a complicated water quality model using the long short-term memory approach. *Journal of Hydrology*, 581, 124432.

- Liu, G. R., & Gu, Y. T. (2005). An introduction to meshfree methods and their programming. Springer Science & Business Media.
- Lv, N., Liang, X., Chen, C., Zhou, Y., Li, J., Wei, H., & Wang, H. (2020). A Long Short-Term Memory Cyclic model With Mutual Information For Hydrology Forecasting: A Case Study in the Xixian Basin. *Advances in Water Resources* .103622.
- Mahar, P. S., & Datta, B. (2000). Identification of pollution sources in transient groundwater systems. *Water Resources Management*, 14(3), 209-227.
- Majumder, P., & Eldho, T. I. (2017). Vectorized simulation of groundwater flow and contaminant transport using analytic element method and random walk particle tracking. *Hydrological Processes*. 31(5), 1144-1160.
- Majumder, P., & Eldho, T. I. (2020). Artificial Neural Network and Grey Wolf Optimizer Based Surrogate Simulation-Optimization Model for Groundwater Remediation. *Water Resource Management*. 34(2), 763-783.
- Matheron, G. (1963). Principles of geostatistics. *Economic geology*, 58(8), 1246-1266.
- Meenal, M., & Eldho, T. I. (2011). Simulation of groundwater flow in unconfined aquifer using meshfree point collocation method. *Engineering Analysis with Boundary Elements*. 35(4), 700-707.
- Mirjalili, S., Mirjalili, S. M., & Lewis, A. (2014). Grey wolf optimizer. *Advances in Engineering Software*. 69, 46-61.
- Mirjalili, S., Mirjalili, S. M., & Hatamlou, A. (2016). Multi-verse optimizer: a nature-inspired algorithm for global optimization. *Neural Computing and Applications*. 27(2), 495-513.
- Morris, M. S., & Thorne, K. S. (1988). Wormholes in spacetime and their use for interstellar travel: A tool for teaching general relativity. *American Journal of Physics*. 56(5), 395-412
- Nikroo, L., Kompani-Zare, M., Sepaskhah, A. R., & Shamsi, S. R. F. (2010). Groundwater depth and elevation interpolation by kriging methods in Mohr Basin of Fars province in Iran. *Environmental monitoring and assessment*, 166(1), 387-407.
- Oliphant, T. E. (2006). A guide to NumPy (Vol. 1, p. 85). USA: Trelgol Publishing.
- Pan, Z., Lu, W., & Chang, Z. (2021). Simultaneous identification of groundwater pollution source spatial-temporal characteristics and hydraulic parameters based on deep regularization neural network-hybrid heuristic algorithm. *Journal of Hydrology*, 126586.

- Panahi, M., Sadhasivam, N., Pourghasemi, H. R., Rezaie, F., & Lee, S. (2020). Spatial prediction of groundwater potential mapping based on convolutional neural network (CNN) and support vector regression (SVR). *Journal of Hydrology*, 588, 125033.
- Pathania, T., Bottacin-Busolin, A., Rastogi, A. K., & Eldho, T. I. (2019). Simulation of groundwater flow in an unconfined sloping aquifer using the element-free Galerkin method. *Water Resources Management*, 33(8), 2827-2845.
- Pathania, T., Eldho, T. I., & Bottacin-Busolin, A. (2020). Optimal design of in-situ bioremediation system using the meshless element-free Galerkin method and particle swarm optimization. *Advances in Water Resources*. 144, 103707.
- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., ... & Chintala, S. (2019). Pytorch: An imperative style, high-performance deep learning library. *Advances in neural information processing systems*, 32, 8026-8037
- Refaeilzadeh, P., Tang, L., & Liu, H. (2009). Cross-validation. *Encyclopedia of database systems*, 5, 532-538.
- Rastogi, A. K. (2007). Numerical groundwater hydrology. Penram International Publication, India.
- Sayed, G. I., Darwish, A., & Hassanien, A. E. (2019). Quantum multiverse optimization algorithm for optimization problems. *Neural Computing and Applications*, 31(7), 2763-2780.
- Singh, R. M., & Datta, B. (2004). Groundwater pollution source identification and simultaneous parameter estimation using pattern matching by artificial neural network. *Environmental Forensics*. 5(3), 143-153.
- Singh, R. M., & Datta, B. (2006). Identification of groundwater pollution sources using GA-based linked simulation optimization model. *Journal of Hydrologic Engineering*. 11(2), 101-109.
- Singh, L. G., Eldho, T. I., & Kumar, A. V. (2016). Coupled groundwater flow and contaminant transport simulation in a confined aquifer using meshfree radial point collocation method (RPCM). *Engineering Analysis with Boundary Elements*, 66, 20-33.
- Sun, A. Y., Painter, S. L., & Wittmeyer, G. W. (2006). A robust approach for iterative contaminant source location and release history recovery. *Journal of Contaminant Hydrology*. 88(3-4), 181-196.
- Sutskever, I., Vinyals, O., & Le, Q. V. (2014). Sequence to sequence learning with neural networks. In *Advances in neural information processing systems* (pp. 3104-3112).
- Tegmark, M. (2004). Barrow, JD Davies, PC Harper, CL, Jr eds. Science and Ultimate Reality Cambridge University Press Cambridge.

- Thomas, A., Majumdar, P., Eldho, T. I., & Rastogi, A. K. (2018). Simulation optimization model for aquifer parameter estimation using coupled meshfree point collocation method and cat swarm optimization. *Engineering Analysis with Boundary Elements*, 91, 60-72.
- Wagner, Brian J. Simultaneous parameter estimation and contaminant source characterization for coupled groundwater flow and contaminant transport modelling. *Journal of Hydrology*, 135, no. 1-4 (1992): 275-303.
- Wunsch, A., Liesch, T., & Broda, S. (2021). Groundwater level forecasting with artificial neural networks: a comparison of long short-term memory (LSTM), convolutional neural networks (CNNs), and non-linear autoregressive networks with exogenous input (NARX). *Hydrology and Earth System Sciences*, 25(3), 1671-1687.
- Xiang, Z., Yan, J., & Demir, I. (2020). A rainfall-runoff model with LSTM-based sequence-to-sequence learning. *Water resources research*, 56(1), e2019WR025326.
- Xing, Z., Qu, R., Zhao, Y., Fu, Q., Ji, Y., & Lu, W. (2019). Identifying the release history of a groundwater contaminant source based on an ensemble surrogate model. *Journal of Hydrology*, 572, 501-516.
- Yadav, B., Ch, S., Mathur, S., & Adamowski, J. (2016). Estimation of in-situ bioremediation system cost using a hybrid Extreme Learning Machine (ELM)-particle swarm optimization approach. *Journal of Hydrologic Engineering*, 543, 373-385.
- Yeh, H. D., Chang, T. H., & Lin, Y. C. (2007). Groundwater contaminant source identification by a hybrid heuristic approach. *Water resources research*, 43(9).
- Zhao, Y., Lu, W., & Xiao, C. (2016). A Kriging surrogate model coupled in simulation–optimization approach for identifying release history of groundwater sources. *Journal of Contaminant Hydrology*, 185, 51-60.
- Zhang, J., Zhu, Y., Zhang, X., Ye, M., & Yang, J. (2018). Developing a Long Short-Term Memory (LSTM) based model for predicting water table depth in agricultural areas. *Journal of Hydrology*, 561, 918-929.
- Zheng, C., & Wang, P. P. (1999). MT3DMS: a modular three-dimensional multispecies transport model for simulation of advection, dispersion, and chemical reactions of contaminants in groundwater systems; documentation and user's guide.